Serial Entrepreneurship in China*

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Abstract

This paper studies entrepreneurship and firm creation through the lens of serial entrepreneurs – entrepreneurs who establish more than one firm. Using data covering the universe of Chinese firms, we document key facts about serial and non-serial entrepreneurs and develop a theory rationalizing how their behavior is shaped by ability, endowments, and capital market frictions. Serial entrepreneurs have higher average productivity and lower return on capital than non-serial entrepreneurs. However, this premium conceals significant heterogeneity, with a majority of serial entrepreneurs underperforming relative to non-serial entrepreneurs. We link these differences to capital market frictions favoring some, but not all, entrepreneurs.

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1 Introduction

The creation of new productive firms is an important engine of growth, especially in emerging economies. While most entrepreneurs start only a single firm during their lifetime, some start and operate a series of firms. This paper studies entrepreneurship and the creation of new firms in China through the lens of these serial entrepreneurs (SEs) and their differences with entrepreneurs who only start a single firm, or non-serial entrepreneurs (non-SEs). These differences are informative about the underlying frictions and dynamics of entrepreneurship and more generally, the environment facing potential entrepreneurs.

In principle, serial entrepreneurs could be selected because of highly persistent skills, i.e., they are individuals good at identifying new business opportunities (Holmes and Schmitz, 1995). Alternatively, their advantage may lie in better access to scarce factors or connections, which help them secure finance and overcome barriers to doing business. If serial entrepreneurship is driven by persistent skills, we expect them to be better than average; if driven by non-skill advantages, we expect them to have lower productivity and to be associated with resource misallocation. Thus, the performance of SEs in terms of total factor productivity (TFP) and assets relative to non-SEs is revealing of the driving forces of entrepreneurship as well as potential frictions entrepreneurs face.

Studying serial entrepreneurs is often constrained by data availability. In this paper, we leverage two unique data sources for Chinese firms: the Business Registry of China and the Inspection Database. These data allow us to document salient facts about entrepreneurship and serial entrepreneurship over the period between 1995-2015 (Registry Data) and measure their performance over the shorter window between 2008-2012 (Inspection Data).

To guide our empirical assessment, we develop a simple two-period model of firm creation in the spirit of Hopenhayn (1992). Each period, potential entrepreneurs receive the option to start a new firm with a stochastic productivity. Operating a firm requires capital, and entrepreneurs can use their own equity or rent capital from banks. Borrowing is limited by collateral constraints: entrepreneurs can collateralize only a fixed fraction of their capital stock. The relationship between firm debt, equity and TFP in our data substantiates the presence of these constraints in China.

The relative productivity of SEs in the model is shaped by a fundamental trade-off between persistence of skills (i.e., productivity across different potential firms for an individual entrepreneur) and heterogeneity in the entrepreneurs' access to credit (i.e., non-skill advantages at starting firms). If skills are sufficiently persistent, SE firms are on average more productive than non-serial firms, and the second SE firm is more productive than the first SE firm. However, if this persistence is sufficiently low and some entrepreneurs have large

non-skill advantages, serial entrepreneurs have lower TFP than non-serial entrepreneurs.

Our empirical evidence is in line with the persistent-skills force dominating: serial entrepreneurs' firms are significantly more productive than non-serial firms in the same (2 digit) industry and the 2nd-SE firm is more productive than the 1st-SE firm. Relative to their peers, the 1st- and 2nd-SE firms are 11% and 21% more productive, respectively. Thus, on average, the evidence for China points toward entrepreneurs becoming SEs for efficiency reasons. However, this behavior conceals substantial heterogeneity among SEs in their productivity that is related to their industry choices.

To study these linkages, we extend our model to incorporate the choice of industry across firms established by a single serial entrepreneur. We focus on a mechanism of learning about industry-specific comparative advantage: if an entrepreneur is relatively successful (unsuccessful) at operating a firm in a particular industry, then a future firm started in the same industry by the same entrepreneur can also be expected to be relatively successful (unsuccessful). This learning induces entrepreneurs to stay in the same industry if the TFP of the first firm is large and switch industry if it is sufficiently low (Jovanovic, 1979).

This threshold behavior interacts with potential heterogeneity in non-skill advantages. Assume that some entrepreneurs are "favored" by having easier access to borrowing. These favored entrepreneurs should have a lower TFP threshold for entry because their preferential access to finance allows them to recover the fixed operating cost at a lower level of TFP. They should also install more capital in the firms they operate, conditional on TFP. Favored entrepreneurs are therefore negatively selected on TFP and positively selected on capital relative to their non-favored counterparts. Thus, favored entrepreneurs should be over-represented among the industry switchers relative to their actual share in the population, which motivates using switching behavior as an indicator for favored entrepreneurs. We find that switchers have, on the one hand, significantly lower TFP and, on the other hand, more capital than stayers and non-serial entrepreneurs. The fact that switchers have lower TFP than non-serial entrepreneurs implies that the entire SE productivity premium is driven by stayers.

Finally, in the presence of financial constraints, our analysis has implications for the decision of serial entrepreneurs to operate their firm concurrently or non-concurrently, i.e., close the old firm when establishing the new firm. Our theory predicts that entrepreneurs with smaller endowments (equity) should close the first firm when the second firm is substantially more productive because of the higher opportunity costs of capital associated with keeping the first firm in operation. This prediction is borne out in the Chinese data.

We consider two additional motives for the choice of industry for the 2nd SE firm: diversification of industry-specific risk and gains from complementarities between firms operating

in related industries. To study diversification, we assume there are industry-specific returns to capital, which creates a hedging motive and preference for industries with a low covariance in returns. As a result, an entrepreneur requires a risk premium in the form of higher productivity to start a second firm in the same industry as the first firm. Empirically, we find that entrepreneurs' industry choices are consistent with a diversification motive: industries with a lower covariance (of return to capital) with the industry of the first firm are chosen more frequently. Moreover, the firms in the industries with low covariance (with the first industry) have lower TFP on average. In addition, conditional on switching industries, entrepreneurs are more likely to choose industries with strong input-output linkages and industries that share the same downstream and/or upstream links.

Our paper contributes to the growing literature that recognizes that entrepreneurship is a "serial" activity. Recent studies document that serial entrepreneurship is common in the US and other countries and highlight the relative importance of entrepreneurial "learning by doing" and ability in understanding serial entrepreneurship.[1] In each case, authors find that serial entrepreneurs are more successful than non-serial entrepreneurs measured in terms of either size (sales), firm longevity, or productivity, which they attribute to the ability of these entrepreneurs and/or faster learning.[2] In common with much of the literature, we find that SEs are on average better. However, this premium conceals significant heterogeneity, with an important subset of serial entrepreneurs underperforming relative to non-SEs, which we link to distortions that favors some, but not all entrepreneurs.

We also contribute to the broader literature on entrepreneurship (e.g., Quadrini (2000), Cagetti and De Nardi (2006), and Tan (2018)) and the effect of distortions and misallocations on economic development and growth (see e.g., Restuccia and Rogerson (2008) and Hsieh and Klenow (2009)). In this context, a country's financial system is seen as critical to the nexus between entrepreneurship and growth (King and Levine (1993a,b)). Especially important is the role of liquidity and financial constraints (Evans and Jovanovic (1989)), which may prevent individuals from starting new firms and lead to a misallocation of resources among currently operating firms (e.g., Erosa (2001), Buera, Kaboski and Shin (2011), Midrigan and Xu (2014), and Moll (2014)). These papers, however, do not analyze serial entrepreneurship.

Finally, in the context of China our study identifies a mix of dynamism and distortion, a

[1]See, for example, Holmes and Schmitz (1990, 1995), Chari, Golosov and Tsyvinsky (2005), and Lafontaine and Shaw (2016) for the US; Westhead and Wright (1998) for the UK; Wagner (2003) for Germany; Hyytinen and Ilmakunnas (2007) for Finland; Amaral, Baptista and Lima (2011), Rocha, Carneiro and Amorim Varum (2015), and Felix, Karmakar and Sedlacek (2021) for Portugal; Shaw and Sørensen (2019) for Denmark; Carbonara, Tran and Santarelli (2020) for Vietnam.

[2]From an empirical point of view, our paper is related to Rocha et al. (2015), Lafontaine and Shaw (2016), Shaw and Sørensen (2019), Shaw and Sørensen (2022), and Felix et al. (2021), who leverage panel data of entrepreneurs for several industrialized countries, including Portugal, US, and Denmark.

more general feature of the Chinese economy. We show that dynamism is manifested in the presence of highly productive serial entrepreneurs while distortions are betrayed by favored individuals and financial frictions. In the literature, the contrast is typically in terms of differences between the state and non-state sector (Song, Storesletten and Zilibotti (2011), Hsieh and Song (2015), Lardy (2019)). Entry by private firms has also been singled out for its important contribution to productivity growth (Brandt, Van Biesebroeck and Zhang (2012)). We find that even within the private sector there are stark differences among serial entrepreneurs in their productivity and size. We link this to preferential access to finance and favoritism. Given the weight of serial entrepreneurs in the Chinese economy, the costs of these distortions in the aggregate are likely significant.

The rest of the paper is organized as follows. Section 2 describes our data and provides an introductory discussion of entrepreneurship and serial entrepreneurship in China. Section 3 lays out a stylized model of serial entrepreneurship. Section 4 confronts the theoretical predictions with our Chinese data on serial entrepreneurship while Section 5 studies two additional motives for the industry choice of the firms of serial entrepreneurs: diversification of industry-specific risk and gains from complementarities between firms operating in related industries. Section 6 concludes.

2 Datasets and a First Look at Serial Entrepreneurs

2.1 Business Registry of China

Our primary data source for analyzing the behavior of serial entrepreneurs is the Business Registry of China Database which is maintained by the State Administration of Industry and Commerce (SAIC). From now on we will refer to it as the *Registry Data*. The Registry Data provides a regularly updated record of all firms that have operated in China since 1949. We have access to a single snapshot of the Registry for the end of 2015, thus providing information on all firms that have ever operated in China since 1949 through 2015.[3]

For each firm in the Registry, we have information on their year of establishment, exit date (if applicable), primary 4-digit industry, location, registered capital (in nominal terms), and investors in the firm and their share. Registered capital is total capital contributed by all investors in the firm and represents a notion of firm's equity.[4] The firm investors are

[3]The data for later years are less accurate. A combination of reform of the Business Registry system in 2014 and evaluation of local government officials based on new firm entry likely contributed to a growing number of shell (fake) companies. Simultaneously, stricter rules regarding firm liquidation that increased these costs likely reduced firm exit. As explained more fully below, most of our analysis uses data through 2012.

[4]A small percentage of unincorporated firms such as township and village enterprises (TVEs) do not have

classified into three groups − individuals, enterprises, or the government. For each investor, we know their total paid-in capital and date of investment into the firm. We supplement data from the Registry in 2015 with information collected by SAIC on changes in the registered capital of firms.[5]

The structure of the dataset. The Registry Data, and thus the information available to us, has the following structure.

- *Operating firms in 2015.* For all firms operating in 2015, the Registry Data provides information on the current ownership of the registered capital in the firm. For each investor in the firm we know the value and their share of registered capital.
- *Firms that went out of business before 2015.* For firms that went out of business in year T, prior to 2015, the Registry Data provides information on the ownership of the registered capital in the firm in year T.

As we discuss later in section 2.3, this organization of the dataset allows us to have a very precise view of the ownership structure of the Chinese firms in our baseline sample. In the analysis that follows, we focus on the post-1995 time period.

2.2 Inspection Database

The Registry Data provides comprehensive ownership information on all firms ever established in China since 1949. To analyze firm performance, we draw on complementary data collected by SAIC for regulatory purposes that provides annual information on firm's assets, liabilities, total sales, total profit, net profit, total taxes, and equity (computed as assets minus liabilities). We will refer to these data as the *Inspection Data*. Responsibility for collection was delegated to local SAIC offices, and originally was done manually. In 2007, information began to be collected digitally, which contributed to a significant increase in its coverage.[6] The inspection system was abolished in 2014 as part of the reform of the business

registered capital. We retain these firms in our sample but drop those incorporated firms with no registered capital information. We also drop firms, comprising 4.5% of the sample, with missing or invalid information on location and industry.

[5]SAIC records in a separate database all changes in firm location, registered capital, and ownership in the form of unstructured texts. We successfully extract and use the data on changes in registered capital of firms over time, which shows that approximately 10% of firms updated their initial registered capital. The ownership information is harder to extract because of the more complicated format of the data reported.

[6]Between 2007 and 2008, the number of prefectures that were not reporting declined from 159 to 98, and by 2012 fell to 81. We drop provinces with an average reporting ratio below 40% over 2008–2012 in the cleaned inspection data, namely Tianjin, Liaoning, Jilin, Heilongjiang, Zhejiang, Jiangxi, Shandong, Hubei, Hunan, Hainan, Yunnan, Tibet (Xizang), and Xinjiang.

registry system, and replaced by an annual self-reporting system. For quality reasons, we restrict our use to 2008-2012.[7]

2.3 Business Registry of China: Ownership Information

Table 1 provides information on the ownership types of firms in China between 1995 and 2015. In order to classify firms into various ownership categories, we identify the largest shareholder in a firm: either an individual, an enterprise, or the government. For a declining fraction of firms − primarily township and village enterprises − shareholder information is not reported. We classify those as *Unreported* in Table 1.

Firms for which the largest investor is not registered are classified as *Unregistered*. In most of these cases, the largest investor is a government entity such as the State-owned Assets Supervision and Administration Commission (SASAC). In the case of unincorporated enterprises—that is, enterprises not formally registered under China's 2004 Corporate Law, including collectively owned and state-owned enterprises—the largest investor may also remain unregistered.

Firms for which an individual is the largest investor are classified as *Individual*. We further disaggregate the *Individual* firms into two groups: (i) Single, if there is only one investor, and (ii) Multiple, if there are multiple investors. Firms for which an enterprise is the largest investor are classified as an *Enterprise*. These are also divided into two groups based on whether the enterprise is the only investor in the firm or not.

As Table 1 illustrates, China has experienced a significant rise in the number of firms operating in the economy − from 2.9 million (column (1) plus (7)) in 1995 to 20.2 million in 2015. Most of this increase is driven by a dramatic increase in firms with individual investors, either single or multiple, with each increasing by more than a factor of 30 between 1995 and 2015. Even though these *Individual* firms are smaller than the *Enterprise* and the *Unregistered* firms, based on their average registered capital as reported in Table 2, the rapid increase in their numbers makes them the main engine of growth in China over this period.

Universe of firms in the analysis. The analysis in this paper is focused on firms for which the largest shareholder is an individual, either single or multiple, as classified in columns (3) and (4) in Table 1. The share of these firms, excluding the *Unreported* firms in column (7), increases from 33.2% of all firms in 1995 to 90.5% in 2015. By 2015, as reported in Appendix Table A-3 which also excludes the *Unreported* firms, these firms held 44.3% of all the registered capital in the economy, up from 10.9% in 1995.

[7] Appendix Tables A-1 and A-2 provide more information on the number of firms and the reporting ratio in the Inspection Data.

Table 1: Firms in China, by Ownership Type, Registry Data, 1995-2015.

Year	Based on largest shareholder						Unreported
	Total	Unregistered	Individual		Enterprise		
			Single	Multiple	Single	Multiple	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
1995	1,421,372	688,395	178,195	293,831	186,939	74,012	1,494,538
1996	1,636,607	736,454	204,913	403,851	197,772	93,617	1,525,302
1997	1,866,526	774,493	236,125	536,225	206,712	112,971	1,484,429
1998	2,135,983	788,668	280,605	720,705	212,321	133,684	1,362,141
1999	2,395,892	785,180	326,414	916,159	217,700	150,439	1,247,664
2000	2,705,485	772,269	383,019	1,164,107	217,730	168,360	1,097,504
2001	3,052,323	744,724	455,228	1,455,771	212,985	183,615	946,172
2002	3,494,935	729,473	552,956	1,803,473	209,346	199,687	825,518
2003	4,045,915	706,566	663,669	2,249,998	205,621	220,061	725,642
2004	4,675,342	683,129	800,310	2,751,166	202,952	237,785	648,374
2005	5,304,289	659,578	931,795	3,252,721	204,279	255,916	578,990
2006	5,932,844	637,995	1,123,203	3,686,789	212,811	272,046	527,334
2007	6,399,212	606,769	1,278,770	4,008,794	221,055	283,824	468,000
2008	6,929,869	582,079	1,432,373	4,391,802	229,541	294,074	438,490
2009	7,763,981	572,724	1,635,190	5,002,348	242,217	311,502	425,791
2010	8,855,082	572,023	1,897,702	5,783,755	262,069	339,533	415,983
2011	10,111,337	580,153	2,207,056	6,668,021	284,175	371,932	411,725
2012	11,273,431	583,293	2,504,773	7,478,657	304,853	401,855	425,778
2013	13,142,739	620,343	3,074,320	8,673,443	333,106	441,527	479,629
2014	16,157,006	667,398	4,149,747	10,445,934	384,595	509,332	521,010
2015	19,608,863	814,277	5,543,067	12,208,189	448,406	594,924	615,954

Notes: Authors' calculations from the Registry Data. Ownership type is identified based on the largest shareholder in a firm: either an individual, an enterprise, or the government.

The structure of the dataset implies that we cannot reconstruct the entire ownership history of each firm since its establishment. However, we believe this is not a serious drawback for the analysis of our baseline sample of *Individual* firms, either single or multiple. For more than 80% of all *Individual* firms, either operating or out of business in 2015, the individual who currently owns the firm or owned it when it went out of business is also the individual who established it. We are able to infer this from the data since in those cases the year in which the firm was established is identical to the year in which the individual acquired the registered capital in the firm.

Table 2: Average Registered Capital, in Millions of Yuans, by Ownership Type, Registry Data, 1995-2015.

Year	Total	Unregistered	Individual		Enterprise	
			Single	Multiple	Single	Multiple
1995	9.50	9.90	2.23	3.67	17.14	27.14
1996	9.09	10.13	2.27	3.50	16.35	24.56
1997	8.74	10.22	2.31	3.34	16.73	23.09
1998	8.91	10.46	2.27	4.90	17.19	22.20
1999	8.91	11.62	2.24	4.40	18.57	22.70
2000	9.10	12.58	2.16	3.98	21.04	28.92
2001	8.65	13.74	2.22	3.22	23.61	29.62
2002	8.23	14.92	2.41	3.05	22.67	31.48
2003	8.44	16.90	2.39	3.36	29.52	31.77
2004	7.70	18.48	2.34	2.95	31.93	28.95
2005	7.41	20.34	2.30	2.90	33.60	29.23
2006	7.29	22.58	2.22	2.89	34.44	30.75
2007	7.62	27.69	2.12	2.87	36.82	33.91
2008	7.59	30.51	2.13	2.80	38.31	36.31
2009	7.56	32.76	2.15	2.90	40.74	38.82
2010	7.66	35.02	2.19	3.10	43.00	42.33
2011	7.72	36.77	2.18	3.31	44.89	45.98
2012	7.77	38.63	2.17	3.41	46.99	49.20
2013	7.60	38.35	2.10	3.50	47.59	52.95
2014	7.78	38.93	2.35	4.04	48.66	56.99
2015	8.24	36.48	2.70	4.64	50.21	63.40

Notes: Authors' calculations from the Registry Data.

2.4 Serial Entrepreneurs in the Business Registry of China

We now provide our definition of entrepreneurs and serial entrepreneurs, describe how we identify firms that belong to serial entrepreneurs, and discuss several preliminary findings regarding serial entrepreneur firms.

2.4.1 Definitions

Entrepreneur. Based on the universe of firms in our analysis, we define an entrepreneur as an individual investor with the largest share in a firm at the time of the firm's establishment or acquired later.

Serial entrepreneur. We define a serial entrepreneur as an investor that is or has been defined as the "Entrepreneur" for more than one firm. In other words, a serial entrepreneur

is an individual that is the largest shareholder, not necessarily concurrently, in at least two firms.

Identifying a serial entrepreneur (SE) firm. We look at the period up through 2015 in order to identify an individual as a serial entrepreneur. Any entrepreneur that established two or more firms by 2015 is classified as serial. Then, using each firm's establishment date, we classify each serial entrepreneurs' firms as first (1st-SE), second (2nd-SE), and so forth. All other firms are classified as firms belonging to individuals that are not serial entrepreneurs, or non-SE firms.[8] In the following analysis, we use *1st-SE* to represent the first firm of SE and *2nd-SE* to denote second and following firms of SE.[9]

Identifying SE firms as concurrent vs. non-concurrent. We define a serial entrepreneur as "concurrent" if she operates two firms simultaneously and "non-concurrent" if the 1st-SE firm is closed down either before or at the time the 2nd-SE firm is started. Note that if the 1st-SE firm was closed down simultaneously with the start of the 2nd-SE firm, both firms would report some operation in the year the 2nd SE firm is started. We therefore identify two SE firms as non-concurrent firms if the firms report positive activity in the Inspection Data for no more than one year over the data sample (2008-2012). Moreover, we classify such firms as non-concurrent during all of their years of operation in the Inspection Data (2008-2012).[10] Conversely, we identify two SE firms as concurrent firms if both firms report positive activity in the Inspection Data for at least two years over the data sample (2008-2012).[11] By this definition, 60.7% of SE firms are operated concurrently.

[8]Our baseline sample includes firms that are in the Inspection Data and must therefore be started by 2012. These firms are classified as serial versus non-serial based on the Registry Data which ends in 2015. This gives entrepreneurs who start their first firm at least three years to start the second firm and still be classified as serial (the median duration for starting the second firm is about three years). This implies that entrepreneurs who start their first firm before 2012 and the second one after 2015 would necessarily be misclassified as non-serial. To mitigate this potential source of error, we carry out a sensitivity analysis in which we restrict attention to entrepreneurs who started their first firm before 2009. This gives them at least six years (i.e., between 2009 and 2015) to start the second firm, which should minimize the potential classification problem.

[9]Following Lafontaine and Shaw (2016), we drop entrepreneurs who established more than twenty firms to exclude the opening of large chain stores in our analysis. This restriction eliminates only 605 entrepreneurs with a total of 200,453 firms, which is very small compared to the remaining 19,114,959 entrepreneurs with 23,108,150 firms.

[10]An implication of this procedure is that in the analysis of concurrent versus non-current firms, we exclude all pairs of 1st- and 2nd-SE firms where the 2nd-SE firm is started after 2012. Since we do not have Inspection data after 2012, we do not know whether 2nd-SE firms started after 2012 operate concurrently or non-concurrently with the 1st-SE firm. Therefore, these firms cannot be classified.

[11]An exception arises when the last reporting year of the 1st-SE firm is 2012 and both firms report positive activity in only that year. In this case, we cannot determine whether the 1st-SE firm exits in 2013.

2.4.2 Empirical Patterns

SE firms. Table 3 highlights the fact that SE firms are a quantitatively significant part of the Chinese economy − they represent about one-third of all firms and hold almost half of all the registered capital. As shown in column (2), the fraction of SE firms increased from 31.1% in 1995 to 34.9% in 2005.[12] We observe similar behavior with respect to registered capital of SE firms which increased from 42.6% in 1995 to a high of 50.1% in 2011 before receding to 47.2% in 2015.

Columns (5) and (6) in Table 3 reveal that the average registered capital in SE firms is about two times larger than in non-SE firms. The fact that SE firms are larger in terms of capital is central for our analysis, and raises the following question: Are they larger because they are more productive in terms of TFP or because they have easier access to finance?

Entry and Exit of SE and Non-SE firms. The changing role over time of SE firms in terms of their number can be a product of differences in either entry or exit patterns, which we report separately by type of firm in Appendix Table A-4.[13]

Entry rates for 1st-SE firms lag those for non-SE firms for most years, but this is more than offset by the higher entry rate for 2nd-SE firms.[14] Exit rates for SE and non-SE firms are fairly similar for both groups and rise up through 2007 before falling afterwards. In summary, the growing role of SE firms over time documented in Table 3 is due to the higher entry rates of 2nd-SEs and larger registered capital of SEs.

Industrial distribution of entrants. Table 4 reports − separately for non-SE, 1st-SE, and 2nd-SE firms − the distribution across industries for new entrants in 2005 and 2010. We report two measures of the distribution of entrants: unconditional and conditional. The unconditional share measures the actual distribution across industries for new entrants. In order to control for the underlying distribution of firms across industries, we compute a conditional share that measures the distribution of entrants relative to the distribution of all firms across industries in the previous year. A measure above (below) one means that firms are more (less) likely to enter that industry relative to the existing industry distribution of firms.

[12]The fraction recedes to 28.4% in 2015. By construction, we likely underestimate the number and role of SE firms in later years because the window for new entrepreneurs to start 2nd firms narrows as the start date of the first firm approaches 2015.

[13]As pointed out above, because of a reform in the registry system, entry (exit) rates in 2014 and 2015 are not comparable with estimates for earlier years, and likely over (under) estimate true rates.

[14]The entry rate is new firms in the current period as a share of all firms in the previous period. The exit rate is the number of firms that exit in the current period as a share of all firms in the current period.

Table 3: Serial Entrepreneur Firms, Registry Data, 1995-2015.

Year	Number of firms		Total registered capital (in trill.)		Mean registered capital (in mill.)	
	Number	SE (%)	Capital	SE (%)	SE	Non-SE
	(1)	(2)	(3)	(4)	(5)	(6)
1995	370,322	31.07	1.12	42.59	4.16	2.53
1996	500,299	31.60	1.46	44.00	4.05	2.38
1997	657,307	32.17	1.85	44.20	3.87	2.32
1998	880,084	32.76	2.34	45.21	3.66	2.16
1999	1,117,632	33.36	2.88	46.25	3.57	2.08
2000	1,423,916	33.90	3.55	47.32	3.48	1.99
2001	1,793,729	34.35	4.40	47.53	3.39	1.96
2002	2,241,116	34.56	5.29	47.59	3.25	1.89
2003	2,800,787	34.78	6.57	47.94	3.23	1.87
2004	3,448,904	34.82	7.97	48.22	3.20	1.83
2005	4,087,156	34.87	9.44	48.29	3.20	1.83
2006	4,720,611	34.71	11.01	48.27	3.24	1.85
2007	5,205,904	34.60	12.64	48.49	3.40	1.91
2008	5,749,121	34.32	14.25	48.80	3.52	1.93
2009	6,569,815	33.89	16.81	49.17	3.71	1.97
2010	7,616,750	33.39	20.71	49.62	4.04	2.06
2011	8,814,062	32.69	25.35	50.07	4.40	2.13
2012	9,923,702	32.00	29.39	49.74	4.60	2.19
2013	11,686,122	30.74	34.95	49.10	4.78	2.20
2014	14,526,748	29.56	49.33	48.25	5.54	2.50
2015	17,598,668	28.37	67.19	47.20	6.35	2.81

Notes: Authors' calculations from the Registry Data. SE stands for Serial Entrepreneur firms. Capital is calculated based on the registered capital in a firm.

Wholesale and Retail Trade, Manufacturing, and Business Services dominate with relatively small differences overall between non-serial and serial entrepreneurs, and between 1st and 2nd firms of SEs in the breakdown. Serial entrepreneurs are slightly more likely to enter Business Services, R&D, and Real Estate compared to non-serial entrepreneurs, three industries that represent a fifth of all firms.

Industry and locational switching of SEs. In Table 5, we examine the changes of serial entrepreneurs in the industry and the location of their first and second firms. Only one third of serial entrepreneurs start their second firm in the same industry, and two thirds

Table 4: Share of Entrants in Different Industries, Non-SE and SE Firms, Registry Data, 2005 and 2010.

Industry	2005						2010					
	Unconditional share			Conditional share			Unconditional share			Conditional share		
	Non-SE	1st-SE	2nd-SE	Non-SE	1st-SE	2nd-SE	Non-SE	1st-SE	2nd-SE	Non-SE	1st-SE	2nd-SE
Agriculture	2.37	2.40	2.14	1.31	1.32	1.18	10.83	7.66	5.39	2.11	1.49	1.05
Mining	0.76	0.95	0.97	1.06	1.31	1.34	0.29	0.41	0.50	0.45	0.62	0.77
Manufacturing	23.56	20.83	23.02	0.77	0.69	0.76	17.13	14.93	17.83	0.69	0.60	0.72
Power	0.42	0.35	0.53	0.83	0.70	1.05	0.16	0.17	0.31	0.36	0.39	0.71
Construction	5.38	5.41	4.78	1.16	1.17	1.03	5.27	5.15	4.69	1.06	1.03	0.94
Wholesale & Retail	34.11	33.94	31.39	1.00	1.00	0.92	35.90	36.26	33.25	1.09	1.10	1.01
Transportation	3.04	3.12	2.88	1.42	1.45	1.34	2.47	2.48	2.26	0.96	0.97	0.88
Accommodation	1.45	1.48	2.14	0.90	0.92	1.33	1.02	1.15	1.60	0.73	0.82	1.14
IT	3.85	3.73	3.28	1.16	1.13	0.99	3.12	3.25	2.89	0.88	0.92	0.82
Finance	0.20	0.27	0.40	0.92	1.26	1.89	0.27	0.49	0.88	0.93	1.67	3.01
Real Estate	2.31	3.13	4.22	0.82	1.11	1.50	2.75	3.96	5.64	0.91	1.30	1.86
Business Services	10.51	11.92	12.01	1.38	1.56	1.57	10.22	12.71	13.25	1.09	1.35	1.41
R&D & Tech Services	6.12	6.89	6.85	1.20	1.35	1.35	6.13	7.09	7.48	1.04	1.20	1.27
Public Facility	0.48	0.48	0.52	0.90	0.90	0.99	0.33	0.38	0.44	0.73	0.82	0.97
Resident Services	3.37	2.96	2.72	1.16	1.02	0.94	2.60	2.32	1.94	0.92	0.82	0.69
Education	0.18	0.21	0.18	1.43	1.64	1.43	0.11	0.12	0.11	0.79	0.92	0.82
Social Work	0.17	0.19	0.24	1.45	1.65	2.02	0.07	0.07	0.11	0.58	0.57	0.87
Entertainment	1.63	1.64	1.64	1.35	1.36	1.35	1.27	1.37	1.42	0.89	0.96	0.99
Public Administration	0.07	0.08	0.07	0.70	0.75	0.69	0.04	0.04	0.01	0.75	0.73	0.27
International Organizations	0.02	0.03	0.03	0.30	0.34	0.41	0.00	0.00	0.00	0.06	0.08	0.00

Notes: Authors' calculations from the Registry Data. For a particular SE group, the unconditional share of an industry in a given year, is the number of new established firms in that industry divided by the total number of all newly established firms. For a particular SE group, the conditional share of each industry is the ratio of unconditional share divided by the fraction of a given industry in the stock of firms one year before.

change to a different industry.[15] In contrast, a majority (73.2%) start their second firm in the same prefecture as their first firm.

Table 5: Geographical and Industry Location, Registry Data, 1995-2015.

	Stayers	Switchers	Stayers (%)	Switchers (%)	Total (%)
Same prefecture	967,671	1,948,496	24.29	48.91	73.21
Different prefecture	311,814	755,551	7.83	18.97	26.79
Total(%)			32.12	67.88	100.00

Notes: Authors' calculations from the Registry Data. Stayers are SE whose I-O industry code of the 1st-SE firm and the 2nd-SE firm are the same. Switchers are SE whose I-O industry code of the 1st-SE firm and the 2nd-SE firm are different.

[15]Industry is defined using the classification in the Chinese 2007 Input-Output table, which has 135 categories. For manufacturing, this reflects activities between the 2-digit and 3-digit SIC level, and for services, activities at the two-digit SIC level. We use the industry breakdown from the I-O table for consistency with later analysis that takes advantage of the I-O table to estimate upstream and downstream linkages.

3 A Model of Entrepreneurship

We now lay out a two-period model of entrepreneurship. Section 3.1 analyzes the decision to start a firm in the first period. Section 3.2 studies the choices in the second period when the entrepreneur may choose to start a second firm and become a serial entrepreneur. Motivated by the evidence in Table 3 that serial entrepreneurs start firms with more capital, our analysis focuses on two key factors that affect serial entrepreneurship and the performance of serial entrepreneurs relative to non-serial entrepreneurs: (i) persistence in entrepreneurial skills; and (ii) heterogeneity in access to finance where some entrepreneurs face less constraints. Sections 3.1-3.2 abstract from industry choice. In Section 3.3, we extend the model to incorporate the choice of industry across firms established by a single entrepreneur, focusing on the role of learning about industry-specific comparative advantage.

3.1 Choices in the First Period

The economy is populated by a fixed set of (potential) entrepreneurs who may choose to operate firms.[16] Firms produce a homogeneous good with decreasing returns to scale. The production function is Cobb-Douglas,

$$y_i = z_i^{1-\eta} \left(k_i^{1-\alpha} n_i^{\alpha}\right)^{\eta} , \tag{1}$$

where y_i is the firm's value added, k_i is the firm's capital stock, n_i is the firm's employment, z_i is the firm's total factor productivity, $\eta \in (0,1)$, and $\alpha \in (0,1)$. There is a fixed cost ν for operating the firm. This has to be paid every period. For simplicity we abstract from capital depreciation and capital adjustment costs.

Firms hire labor at a constant wage rate w. Entrepreneurs finance capital $k = e + b$ through equity e and debt b. Banks offer loans at rate R but are not willing to lend more than a share $1 - 1/\lambda$ of installed capital, where $\lambda \geq 1$. This constrains the installed capital stock to $k \leq \lambda e$. One can also deposit equity in banks at the same rate R.

The entrepreneur starts with equity $e \geq 0$ and an opportunity to operate (or start) one firm with potential TFP z. Note that the first-period problem is entirely static: it is optimal to operate a firm if and only if the first-period profits are positive. The entrepreneur therefore decides to operate the firm if the TFP z draw is higher than a threshold value $z^*(e)$ which is weakly decreasing in the entrepreneur's equity e. Conditional on having chosen to operate,

[16] In 77% of the firms in our sample, the largest shareholder is also the the firm's Legal Representative – the key person that has the legal authority to interact with Chinese officials, execute the powers and duties of the company, and bear the legal responsibility for the actions of the firm. Therefore, the largest investors are actively involved in the management and operation of their firms.

the firm's objective is then given by

$$\begin{aligned}\Pi &= \max_{k,n,b}\{y - wn - Rb\} \\ &\quad \text{subject to} \\ b &\leq (\lambda - 1)\, e, \qquad k = e + b \geq 0.\end{aligned}$$

An entrepreneur is either constrained, thus investing all his/her equity in the firm and choosing $k = \lambda e$ and $n = \arg\max_n \left\{ z^{1-\eta} \left((\lambda e)^{1-\alpha} n^{\alpha} \right)^{\eta} - wn \right\}$, or unconstrained and choosing $(k, n) = \arg\max_{k,n} \{y - wn - Rk\}$. We summarize the entry decision and the optimal allocation in the following proposition.

Proposition 1. *Consider an entrepreneur with equity e and an option to operate one firm with TFP z. The entrepreneur will operate the firm if and only if TFP is sufficiently large, $z \geq z^*(e)$, where the entry threshold is given by*

$$z^*(e) = \begin{cases} \left(\frac{\nu + R\lambda e}{1-\alpha\eta}\right)^{\frac{1-\alpha\eta}{1-\eta}} (\lambda e)^{-\frac{(1-\alpha)\eta}{1-\eta}} \left(\frac{w}{\alpha\eta}\right)^{\frac{\alpha\eta}{1-\eta}} & e < zk^*/\lambda \\ z^* & e \geq zk^*/\lambda \end{cases}, \tag{2}$$

where

$$\begin{aligned} k^* &\equiv \left(\frac{(1-\alpha)\,\eta}{R}\right)^{\frac{1-\alpha\eta}{1-\eta}} \left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\eta}} \\ z^* &\equiv \frac{\eta}{1-\eta}\frac{1-\alpha}{R}\frac{\nu}{k^*}. \end{aligned}$$

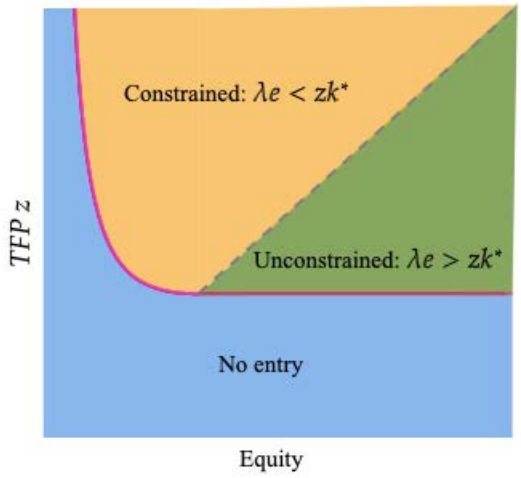
Moreover, for operating firms the optimal installed capital, debt, and debt-equity ratio of the firm are given by the functions

$$\begin{aligned} K^*(z,e) &= \begin{cases} \lambda e & \text{if } \lambda e < zk^* \\ zk^* & \text{if } \lambda e \geq zk^* \end{cases} \qquad (3) \\ B^*(z,e) &= \begin{cases} (\lambda - 1)\, e & \text{if } \lambda e < zk^* \\ zk^* - e & \text{if } \lambda e \geq zk^* \end{cases} \\ \frac{b}{e} = \frac{K^*(z,e) - e}{e} &= \begin{cases} \lambda - 1 & \text{if } \lambda e < zk^* \\ \frac{zk^*}{e} - 1 & \text{if } \lambda e \geq zk^*. \end{cases} \end{aligned}$$

Proof. The proof is in Appendix B.1. □

Figure 1 illustrates how the combination of e and z fall into three distinct regimes: no entry, unconstrained, and constrained. The solid graph $z^*(e)$ marks the indifference between

Figure 1: Entry Decision.



Notes: The figure shows the entry decision for an entrepreneur with equity e and the option to run a firm with TFP z.

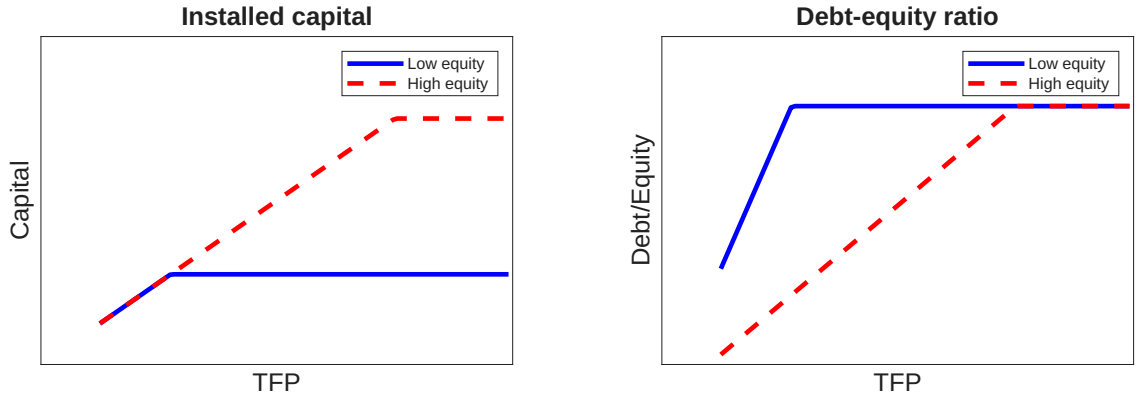
entry and no entry. The entrepreneur chooses to operate the firm if and only if TFP is sufficiently large, $z \geq z^*(e)$. The function $z^*(e)$ is falling in the level of equity e, implying that a larger e is associated with a (weakly) lower threshold $z^*(e)$. The reason is that when e is low, the maximum capital that can be installed is also low. This in turn increases the need for a large TFP in order to recover the fixed cost.

Proposition 1 implies the following theoretical predictions for the relationship between equity, capital, debt, and TFP when firms face financial frictions (i) capital is monotone increasing in TFP and constant for sufficiently high TFP when the financial constraint is binding, (ii) capital is monotone increasing in equity, and (iii) the debt-equity ratio is monotone increasing in TFP and constant for sufficiently high TFP when the constraint is binding. These patterns are illustrated in Figure 2.

3.2 Choices in the Second Period: Becoming a Serial Entrepreneur

Consider now the entrepreneur's choices in the second period. Entrepreneurs consume at the end of the second period, so all potential profits from the first period are saved for the second period. At the beginning of the second period an entrepreneur has equity e and receives a stochastic draw z_2 for a potential new firm. In addition, those entrepreneurs who started and operated a firm in the first period may continue to do so in the second period. The productivity z_1 of the first firm is constant over time. An entrepreneur may choose to operate zero, one, or two firms in the second period. Recall that the entrepreneur can move capital and workers between firms at no cost. We also assume that entrepreneurial effort is

Figure 2: Capital and Debt-Equity Ratio.



Notes: The figure shows the installed capital (left panel) and debt-equity ratio (right panel) for an entrepreneur operating one firm as a function of TFP z for two levels of initial equity.

not a rival input in production.[17]

We analyze separately the two key factors that may affect the performance of serial entrepreneurs: (i) persistence in entrepreneurial skills, and (ii) heterogeneity in access to finance.

3.2.1 Persistence in Entrepreneurial Skills

We capture the persistence in entrepreneurial skills across firms started by an entrepreneur i by allowing the TFP of potential firms drawn in the first and second period to be related via an AR(1) process,

$$z_{i2} = \rho z_{i1} + \varepsilon_i, \tag{4}$$

where the parameter $\rho \in [0, 1]$ captures the autocorrelation of an entrepreneur's TFP. The random component ε_i is i.i.d. and drawn from a symmetric distribution with $E(\varepsilon) = 0$ and c.d.f. F_ε.

No Financial Frictions. To develop the intuition for our results, we initially analyze an environment without any financial constraints, i.e., when $\lambda \to \infty$. From Proposition 1, the optimal decision for the entrepreneur will be to open a firm in the second period if and only if $z_2 \geq z^*$. The following proposition characterizes the average TFP for serial versus non-serial entrepreneurs when there are no financial frictions.

Proposition 2. *Suppose there are no financial frictions ($\lambda \to \infty$). Then,*

[17] This assumption prevents stacking the cards in favor of finding that SE firms are productive than non-SE firms. If entrepreneurial inputs were rival, the measured TFP of (concurrently run) SE firms would be larger in order to overcome a larger span of control across firms.

1. *If TFP draws are persistent ($\rho > 0$), both the 1st-SE and the 2nd-SE will have a larger expected TFP than non-serial entrepreneurs.*

2. *If ρ is sufficiently large, the 2nd-SE will on average have a higher TFP than the 1st-SE.*

3. *If $\rho = 0$, non-serial and serial firms have the same expected TFP.*

Proof. The proof is in Appendix B.2.

□

When the draws are positively correlated, the TFP for the firms of an entrepreneur who obtains two draws in succession above the common threshold z^* must first-order stochastic dominate the TFP for firms of entrepreneurs who obtained one draw above and one below the threshold. Note that the left tail of the realized distribution of ε is truncated from below. The reason is that the entrepreneur would enter with the 2nd firm if and only if $z_2 = \rho z_1 + \varepsilon \geq z^*$. Given z_1, this condition imposes a lower bound on ε that will be realized. If $\rho = 1$, the expected *potential* TFP for the 2nd firm is $z_2 = z_1 + E(\varepsilon) = z_1$. Because the entrepreneur would operate the firm only if the realization of ε were sufficiently large, there is positive selection for the 2nd-SE firm. Therefore, conditional on entry, the realized TFP for 2nd-SE firms must have an expected TFP larger than z_1. Finally, when TFP draws are i.i.d. over time ($\rho = 0$), and potential entrepreneurs are otherwise ex-ante identical, there is no information in being a SE. Therefore, SE and non-SE firms must have the same expected TFP.

Financial Frictions. Consider now the choices for entrepreneurs who face financial frictions in the form of a collateral constraint on borrowing ($\lambda < \infty$). Proposition 3 characterizes the 2nd-period entry decisions for an entrepreneur who operated a firm in the first period.

Proposition 3. *Consider the 2nd-period choice of an entrepreneur who operated a firm with TFP z_1 in the first period. The entrepreneur has equity e and a potential 2nd firm with TFP z_2. The entrepreneur will enter, operate the 2nd firm, and become a Serial Entrepreneur if and only if z_2 is sufficiently large, $z_2 \geq Z(z_1, e)$, where the threshold function Z is given by*

$$Z(z_1, e) = \begin{cases} z^* & when \quad \lambda e \geq (z_2 + z_1) k^* \\ \bar{Z}(z_1, e) & when \quad \lambda e \in [z_1 k^*, (z_2 + z_1) k^*) \\ \underline{Z}(z_1, e) & when \quad \lambda e < z_1 k^*, \end{cases}$$

and

$$\begin{aligned}\bar{Z}(z_1,e) &= \left(1+\frac{1-\eta}{1-\alpha\eta}\left(\frac{z_1k^*}{\lambda e}-1\right)+\frac{(1-\alpha)\eta}{1-\alpha\eta}\frac{1}{\lambda e}\frac{v}{R}\right)^{\frac{1-\alpha\eta}{1-\eta}}\frac{\lambda e}{k^*}-z_1\\ \underline{Z}(z_1,e) &= \left((z_1)^{\frac{1-\eta}{1-\alpha\eta}}+\left(\frac{w}{\alpha\eta}\right)^{\frac{\alpha\eta}{1-\alpha\eta}}\frac{\nu}{1-\alpha\eta}(\lambda e)^{-\frac{(1-\alpha)\eta}{1-\alpha\eta}}\right)^{\frac{1-\alpha\eta}{1-\eta}}-z_1.\end{aligned}$$

The function Z satisfies $Z(z_1,e)\leq z_1$ and is monotone increasing in z_1 and monotone falling in e.

Proof. The proof is in Appendix B.3. □

The TFP threshold for entry of the 2nd firm is now a function $Z(z_1,e)$ of both the TFP of the existing firm, z_1, and equity e. Naturally, the threshold is falling in e because more equity allows the firm to invest in capital closer to the optimal firm size. Moreover, the threshold is increasing in z_1 because a larger productivity of the first firm increases the opportunity cost of equity associated with opening a second firm. To see this, recall that investments are assumed to be fully reversible so the entrepreneur would equate the marginal product of capital across firms if they operate both firms concurrently. The opportunity cost of equity is lower when equity is more abundant. Entrepreneurs with more equity are therefore more likely to start the second firm.

The following corollary characterizes the debt and installed capital in each SE firm.[18] For convenience we relabel the two (potential) firms as h and l, where h indicates the firm with high TFP and l the firm with low TFP ($z_h\geq z_l$). This is without loss of generality because the birth date of each firm is irrelevant as the fixed cost of operation must be paid every period and capital is fully reversible.

Corollary 1. *Consider an entrepreneur who has the option to operate two firms with TFP $z_h\geq z_l$. The installed capital in firms h and l are given by*

$$K_h(z_h,z_l,e)=\begin{cases}\lambda e & when \quad z_l<Z(z_h,e) \ \ and\ \lambda e<z_hk^*\\ z_hk^* & when \quad z_l<Z(z_h,e) \ \ and\ \lambda e\geq z_hk^*\\ \frac{z_h}{z_h+z_l}\lambda e & when \quad z_l\geq Z(z_h,e) \ \ and\ \lambda e<(z_l+z_h)k^*\\ z_hk^* & when \quad z_l\geq \bar{Z}(z_h,e) \ \ and\ \lambda e\geq(z_l+z_h)k^*\end{cases},$$

$$K_l(z_h,z_l,e)=\begin{cases}0 & when \quad z_l<Z(z_h,e)\\ \frac{z_l}{z_h+z_l}\lambda e & when \quad z_l\geq Z(z_h,e) \ \ and\ \lambda e<(z_l+z_h)k^*\\ z_lk^* & when \quad z_l\geq \bar{Z}(z_h,e) \ \ and\ \lambda e\geq(z_l+z_h)k^*\end{cases}.$$

[18] The proof of Corollary 1 follows directly from the proof of Proposition 3.

The total debt of the entrepreneur is given by

$$B\left(z_h, z_l, e\right) = \begin{cases} \left(\lambda - 1\right) e & when \quad z_l < Z\left(z_h, e\right) \ \ and \ \lambda e < z_h k^* \\ z_h k^* - e & when \quad z_l < Z\left(z_h, e\right) \ \ and \ \lambda e \geq z_h k^* \\ \left(\lambda - 1\right) e & when \quad z_l \geq Z\left(z_h, e\right) \ \ and \ \lambda e < \left(z_l + z_h\right) k^* \\ \left(z_h + z_l\right) k^* - e & when \quad z_l \geq \bar{Z}\left(z_h, e\right) \ \ and \ \lambda e \geq \left(z_l + z_h\right) k^* \end{cases}.$$

Corollary 1 characterizes the allocation of capital and debt for each firm of a serial entrepreneur. Both capital and the debt-equity ratio are increasing in the TFP of each firm and capital (in each firm) is increasing in equity.

Financial frictions complicate the predictions of TFP for serial versus non-serial entrepreneurs relative to the complete-market case studied in Proposition 2. The reason is that the level of equity – and in the second period also the TFP of the first firm – now matter for the selection into serial entrepreneurship because these variables affect the TFP thresholds $z^*(e)$ and $Z(z_1, e)$. More equity generally lowers the thresholds, making entry more likely. This gives rise to negative selection on TFP.

TFP in the first firm z_1 affects the probability of entry in the second period and the selection of the second SE firm through three channels.vFirst, a larger z_1 increases the expected realization of z_2 whenever $\rho > 0$. Proposition 2 showed that absent financial frictions, this effect makes entry of 2nd SE more likely and increases the TFP of the 2nd SE firm. This force is also present with financial frictions. However, a larger z_1 implies that the 1st-SE firm generates more profits in the first period. This in turn increases the equity available to the entrepreneur at the beginning of period 2 when z_2 is realized. *Ceteris paribus*, this second channel tends to lower the TFP threshold for the second firm and therefore works in favor of increasing the probability of entry and lowering the expected TFP of the 2nd SE. Third, a larger z_1 increases the opportunity cost of equity for the second firm. Namely, the more productive the first firm is (i.e., larger z_1), the higher the return on equity in the first firm. Therefore, it becomes more costly to tie up this capital in a second firm and drain capital from the first firm. This third channel suggests that a larger z_1 will reduce the probability of entry for the 2nd SE and create positive selection on TFP for the 2nd SE firm. We will return to this third channel in Section 3.2.3 below.

This discussion shows that the predictions for TFP of serial versus non-serial entrepreneurs depend on the persistence of TFP, ρ, the length of the first period, and the initial joint distribution of equity and TFP. In particular, the fundamental trade-off between persistence of skills and heterogeneity in the entrepreneurs' access to credit (i.e., non-skill advantages at starting firms) is shaped by these variables. To make progress on studying the performance of serial entrepreneurs, we will now explore two alternative restrictions on the fundamen-

tals of the model. This will allow us to obtain sharp predictions about TFP and, in turn, help determine whether the persistence of skills view or the non-skill advantages view will dominate.

We first focus on the productivity persistence view. The productivity persistence channel would be weakened if initial TFP and equity were negatively correlated or the length of a period was relatively long. With a negatively correlated initial TFP and equity, the set of entrepreneurs entering in the first period could potentially be dominated by (negatively selected) low-productive entrepreneurs with abundant equity. With a long initial period, entrepreneurs who operate a firm in the first period (and who are candidates to become serial) would have time to accumulate more equity and would hence be subject to more negative selection for the second firm.

To get the productivity persistence view to dominate, it is necessary to ensure that the negative selection associated with more equity does not undo the positive selection associated with productivity persistence. We provide a set of sufficient conditions for this to happen by imposing two restrictions: (1) initial equity and initial TFP must be positively correlated; and (2) the productivity persistence must be sufficiently large. We state the first restriction formally as follows,

Restriction 1. *Assume that initial equity is monotone increasing in the initial TFP draw z_1 in the first period.*

Our earlier proof of Proposition 2 relied on all potential entrepreneurs having the same TFP threshold $z^* \in \mathbb{R}$ for starting firms. Restriction 1 guarantees that the entry condition for the 1st-SE, i.e., that z_1 exceeds the TFP threshold function $z^*(e)$, can be replaced by the condition $z_1 \geq z^{**}$, where $z^{**} \geq z^*(e) \geq z^*$. Namely, there exists a TFP level $z^{**} < \infty$ so that an entrepreneur in period 1 will enter if and only if her TFP draw z_1 satisfies $z_1 \geq z^{**}$. From Restriction 1, any entrepreneur with $z_1 > z^{**}$ has at least as much equity as entrepreneurs with $z_1 = z^{**}$. They will choose to enter because they have both more equity and larger TFP than entrepreneurs with $z_1 = z^{**}$, who are indifferent between entering and not entering. Conversely, no entrepreneur with TFP $z_1 < z^{**}$ will enter because they have lower TFP and less equity than those with $z_1 = z^{**}$. With a common entry threshold for TFP z^{**}, we can use the same proof strategy as for Proposition 2 to show that SEs are more productive than non-SEs if ρ is sufficiently large.

Empirical support for Restriction 1 is provided by the bin scatter plot of equity against TFP for newly established firms reported in Appendix Figure A-1. We can now state the main prediction about TFP in the presence of financial frictions.

Proposition 4. *Suppose there are financial frictions ($\lambda < \infty$) and that Restriction 1 holds. Consider an entrepreneur who operated a firm in the first period with TFP z_1, and who has the option to start a potential firm in period 2 with TFP z_2. If the persistence of TFP draws ρ is sufficiently large, both the 1st-SE and the 2nd-SE will have a larger expected TFP than non-serial entrepreneurs and the 2nd-SE firm will on average have a higher TFP than the 1st-SE firm.*

Proof. The proof is in Appendix B.4. □

In summary, the predictions of Proposition 2 hold up in the presence of financial frictions provided that the persistence ρ is sufficiently large and that Restriction 1 holds.

3.2.2 Heterogeneity in Access to Finance

Having analyzed the role of persistence in skills for serial entrepreneurship, we now turn to the role of heterogeneity in access to finance. In order for these differences to dominate in the selection of serial entrepreneurs, the negative selection associated with equity and access to finance must dominate the positive selection associated with productivity persistence. Sufficient conditions for this to hold therefore would be that the persistence of TFP and the correlation of initial TFP and initial equity are sufficiently low. To simplify the exposition we eliminate any role for persistence in productivity by assuming that $\rho = 0$. We also simplify the exposition by assuming that financial conditions (initial equity, collateral constraints, and interest rates) are independent of TFP draws.

Incorporating heterogeneity in access to finance in a model of China's economy is motivated by a large literature emphasizing substantial differences across firms in the extent of the financial frictions they face (see, e.g., Hsieh and Klenow (2009) and Song et al. (2011)). Entrepreneurs may therefore differ ex ante in their access to borrowing through the tightness of their borrowing constraint. To model this possibility, assume that potential entrepreneurs are of two types, A and B, and that type B individuals (favored) have a more relaxed collateral constraint than type A individuals (non-favored): $\lambda_B > \lambda_A$. There are no other differences between favored and non-favored individuals. In particular, both groups have the same distribution of (potential) TFP.

Propositions 1 and 3 imply that entrepreneurs of type B have a lower TFP threshold because they have easier access to borrowing (higher λ_B). A lower TFP threshold makes them more likely to enter each period. As a result, the favored (type-B) individuals will be over-represented among the serial entrepreneurs relative to their share in the population of entrepreneurs who enter in the first period. Since the favored entrepreneurs have a lower

TFP threshold and $\rho = 0$, serial entrepreneurs will on average have *lower* TFP than non-SEs.[19] However, if λ_B is sufficiently large, the favored entrepreneurs will have more capital, conditional on TFP.

We state these results in the following proposition.

Proposition 5. *If $\rho = 0$ and favored entrepreneurs have a higher λ relative to non-favored entrepreneurs, the favored entrepreneurs will be over-represented among SEs. SE firms will then on average have lower TFP than non-SE firms. Moreover, SE firms will have a larger capital stock conditional on TFP.*

Persistence and heterogeneity in access to finance support contrasting outcomes for the TFP of SE firms relative to non-SE firms. The data will determine which force is dominant.

3.2.3 Operating SE Firms Concurrently or Non-concurrently

When starting their 2nd firm serial entrepreneurs must decide whether to operate their firms concurrently or to close the first firm and just operate the second one. Our model is well suited to study this decision, which is revealing of the financial frictions facing entrepreneurs.

Consider the case when it is optimal for the entrepreneur to start the second firm (and become a serial entrepreneur). Recall that both the 1st-SE and 2nd-SE firm must pay the fixed operating cost ν if they operate in the second period. Moreover, if the two firms operate concurrently, the marginal product of capital will be equated across firms. Since the optimal size of a firm is increasing in its TFP, it follows that the opportunity cost of equity associated with operating the 1st-SE in the second period is increasing in the TFP of the 2nd-SE. Therefore, if equity is sufficiently scarce and the 2nd-SE firm is sufficiently productive, it is optimal to close the 1st-SE firm and allocate all funds to the most productive firm. This effect is stronger the less equity the entrepreneur has in the beginning of period 2.[20]

The optimal choice, which follows from Proposition 3, is characterized in the following proposition.[21]

[19] To see this, note that for any state variable (z_1, e_2), the threshold function Z is monotone decreasing in the collateral parameter λ. A lower threshold is associated with a higher probability of becoming a SE and lower realized TFP for those firms. Therefore, entrepreneurs who are allowed to borrow more for a given equity (larger λ), would be more likely to become SEs and would tend to have a lower TFP.

[20] A similar effect is present if the initial firm is very productive, in which case the opportunity cost of equity would be determined by a potential and less productive 2nd-SE firm. However, if a potential second firm is not sufficiently productive, it would not get started and the entrepreneur would be recorded as a non-serial entrepreneur.

[21] This is further illustrated in Appendix B.5.

Proposition 6. *Consider an entrepreneur with equity e who operated a firm in the first period with TFP z_1 and who has a draw of a potential firm in period 2 with TFP sufficiently large to become a SE, $z_2 \geq Z(z_1, e)$. The entrepreneur will operate the two firms concurrently if z_1 is sufficiently close to z_2. The number of firms operated concurrently by an entrepreneur is monotone increasing in equity and monotone decreasing in the absolute TFP difference $|z_2 - z_1|$.*

Proof. From Proposition 3, the two firms will be operated concurrently if both $z_2 \geq Z(z_1, e)$ and $z_1 \geq Z(z_2, e)$. The inequality condition follows immediately as a necessary and sufficient condition for concurrent operation. □

3.3 Learning Industry-Specific Abilities

So far, we have abstracted from the industry choice of serial entrepreneurs. Slightly more than two thirds of all serial entrepreneurs switch industries when starting their 2nd SE firm. As we shall see, this decision is informative about the factors driving serial entrepreneurship. To incorporate industry choice into our analysis, we assume that there are several industries $s \in S = \{1, 2, ..., N\}$ in the economy. Moreover, an entrepreneur's ability differs across industries. Following Jovanovic (1979), the decision to enter an industry is made before observing the ability. When entering an initial industry s, the potential entrepreneur obtains an initial TFP draw z_{1s}. A firm established in a particular industry must remain in that industry. After the first period, the entrepreneur is free to start a second firm (2nd-SE) in any industry $s' \in S$.[22]

We assume that the persistence of TFP across firms is higher for firms in the same industry than for different industries.[23] In particular, if the entrepreneur chooses to make a TFP draw in the same industry s as the first firm, the persistence of TFP is ρ, while for a TFP draw in a different industry $s' \neq s$, the persistence is $\underline{\rho}$, where $0 < \underline{\rho} < \rho$. Therefore,

$$z_2 = \begin{cases} \rho z_{1s} + (1-\rho)\bar{z} + \varepsilon & \text{if stays in same industry } s \\ \underline{\rho} z_{1s} + (1-\underline{\rho})\bar{z} + \varepsilon & \text{if switches to different industry } s' \in S \backslash s. \end{cases} \tag{5}$$

Moreover, we set $\bar{z} \equiv E\{z_1\}$ to ensure that the expected TFP of *potential* draws of z_2 is the same for switchers and stayers. Thus, any difference in realized outcomes must be due to

[22]This section abstracts from the choice of running firms concurrently or not. All results in this section hold true regardless of whether the firms are concurrent or non-concurrent. In Section 5 we study sector choice in settings where it matters whether or not firms are operated concurrently.

[23]In a previous version of this paper we generalized this process to allow for an intermediate group of industries, dividing industries into "same," "similar," and "distant." All the qualitative properties go through with the current simpler setup.

selection into staying or switching.

All industries are ex-ante identical, with the same distributions for z_1 and ε. The optimal industry choice after the first period is now straightforward: the entrepreneur stays in the same industry as the first firm if and only if the TFP of the 1st-SE firm z_{1s} is larger than the switching threshold $\bar{z}$. The reason is that the entrepreneur prefers high persistence − which she can obtain by staying in the same industry − if and only if the initial draw z_1 is above the average first-period draw, $E\{z_1\}$. This threshold behavior in industry choice induces positive selection for the entrepreneurs who remain in the same industry and negative selection for those who search for potential firms in different industries. Therefore, TFP for serial entrepreneurs locating two firms in the same industry will be larger than TFP for switchers. This captures the spirit of the Jovanovic (1979) learning model. We summarize this decision in the following proposition.

Proposition 7. *Consider an entrepreneur who operated a firm in industry s with TFP z_{1s} in the first period.*

- ***Part A: Industry choice**. If TFP of the 1st-SE firm is above a switching threshold $z_{1s} \geq \bar{z} = E\{z_1\}$, the entrepreneur remains in the same industry s. Otherwise, the entrepreneur switches to a different industry $s' \neq s$.*

- ***Part B: TFP**. Average TFP is higher for entrepreneurs who stay in the same industry than for those who switch industries. This holds true for both the 1st-SE and the 2nd-SE firms.*

Proof. Remaining in the industry implies a larger z_{2s} for all realizations of ε if and only if $z_{1s} > E\{z_1\}$. Moreover, z_{2s} is invariant to ε if $z_{1s} = E\{z_1\}$. This proves Part A. Part B then follows from $\rho > \underline{\rho} > 0$, and z_{1s} being strictly larger for entrepreneurs who stay in the initial industry. □

4 Empirical Analysis

Our model generates empirical predictions relating to: (i) the relationship between productivity, debt, and equity (Corollary 1); (ii) the performance of serial versus non-serial entrepreneurs (Propositions 4 and 5) and concurrent versus non-concurrent SE firms (Proposition 6); and (iii) productivity and the decision of serial entrepreneurs to switch industries (Proposition 7). We examine each in turn.

4.1 Measuring Firm TFP

We measure firm TFP relative to other firms in the same I-O industry, province, and year. This requires data on value added, capital, and labor. We use firm revenue and firm assets from the Inspection Data as measures of value added and capital, respectively.[24] However, the Inspection Data do not have information on labor input. To overcome this shortcoming we assume that all firms in the same I-O industry, province, and year cell pay workers the same wage rate w. Assuming competitive labor markets, firm optimization equates the marginal product of labor across firms. From the firm's first-order condition, labor input can be expressed as $n_i = y_i \alpha\eta / w$. Combining this expression with the production function in equation (1), TFP of firm i relative to the average TFP $E\{z\}$ in a particular province-industry-year cell can be expressed as

$$\frac{z_i}{E\{z\}} = \frac{y_i^{\frac{1-\alpha\eta}{1-\eta}} k_i^{-\frac{(1-\alpha)\eta}{1-\eta}}}{\sum_j y_j^{\frac{1-\alpha\eta}{1-\eta}} k_j^{-\frac{(1-\alpha)\eta}{1-\eta}}}. \tag{6}$$

We assume a return to scale of $\eta = 0.85$ and, following Hsieh and Klenow (2009), we set the labor income share $\alpha\eta$ for each industry equal to the equivalent share in U.S. data.[25] Before computing TFP, we also drop firms with zero assets and revenue. [26] Finally, when reporting the facts on TFP, we drop the bottom and top 1 percentiles of computed TFP.

4.2 Productivity, Debt and Equity

Using the Inspection Data for 2008-2012, we first divide our sample of firms into four quartiles of equity (with the first quartile containing the 25% of firms with the lowest equity, etc.) Next, we sort firms in each quartile on the basis of their TFP into twenty ventiles. For each ventile we calculate average TFP, assets, and debt-equity ratio and present these in scatter

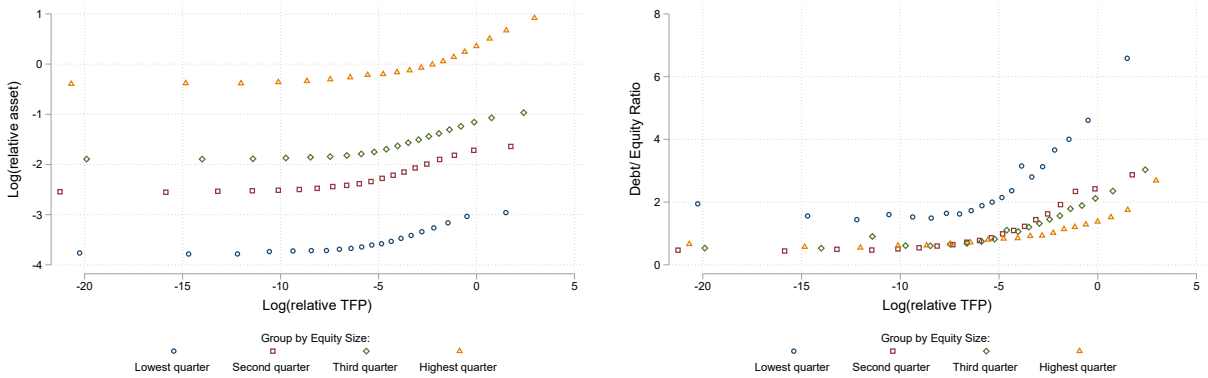
[24] We impute value added by multiplying firm revenue by the ratio of value added to firm revenue at the 2-digit industry level, computed from the 2008 Economic Census.

[25] In Appendix D we discuss alternative methods for estimating the labor share from Chinese data. Using evidence from the Chinese 2007 Input-Output Tables and the 2008 Chinese Enterprise Census Data we argue that U.S. industry-specific labor-income shares are a good estimate for labor elasticities in Chinese data while being arguably less subject to measurement issues.

[26] We drop firms with missing values for assets, revenue, or liabilities. The inspection data also suffer from a unit reporting problem: the correct unit is 10,000 Yuan, but some firms report values in 1 Yuan. To address this, we calculate equity as assets minus liabilities, since registered capital should roughly correspond to equity. We assume the reporting unit is correct if the ratio of equity to registered capital lies between 0.1 and 100. If the ratio falls between 1,000 and 1,000,000, we treat it as a unit error and adjust accordingly. Firms with ratios outside these ranges are dropped.

plots.[27]

Figure 3: Capital, Debt-Equity Ratio, and Relative TFP, Inspection Data, 2008-2012.



(a) Capital and TFP

(b) Debt-Equity Ratio and TFP

Notes: Authors' calculations from the Inspection Data. The left panel shows the relationship between assets and TFP. The right panel shows the relationship between the debt-equity ratios and TFP. Firms are sorted into four quartiles according to their equity. For each equity quartile, firms are ranked on TFP and sorted into twenty ventiles. The left panel plots log average assets against log average TFP for firms in each ventile. The right panel plots log average debt over average equity against TFP. All variables are computed relative to their averages of all firms in the same province-industry-year cell.

The left panel in Figure 3 and column (1) in Appendix Table A-5 show that firms with higher TFP have more assets and are thus larger, especially for firms with TFP above the median. However, note that for sufficiently high TFP, assets are constant in TFP. This pattern is consistent with a binding financial constraint. Assets are also increasing in equity, conditional on TFP. The left panel in Figure 3 verifies that the entire profile of assets shifts upwards as we move from lower to higher equity quartiles.

Finally, the right panel in Figure 3 and column (2) in Appendix Table A-5 show that the debt-equity ratio is increasing in TFP and decreasing in equity − the debt-equity profile is increasing with TFP within each quartile of equity while, as we move from lower to higher equity quartiles, the entire debt-equity profile shifts downward and the slope with respect to TFP becomes flatter.

These patterns are consistent with the relationship among equity, capital, debt, and TFP in our stylized collateral-constraint model of financial frictions − discussed in Proposition 1 and illustrated in Figure 2. In particular, the evidence suggests that the financial constraints are binding for firms with high TFP. We conclude that financial frictions are a quantitatively important feature of the environment facing entrepreneurs in our data.

[27] All variables are computed relative to that variable's average for firms in the relevant province-industry-year cell. Appendix 4.1 provides details on the procedure.

4.3 TFP of Serial Entrepreneur Firms

4.3.1 Serial versus Non-serial Firms

Our key predictions relating to the performance of serial versus non-serial entrepreneurs were provided by Propositions 4 and 5. Table 6 uses data from the 2008-2012 Inspection Data to estimate the levels of TFP, assets, revenue, and equity of 1st-SE and 2nd-SE firms relative to non-SE firms. All variables, except age, are computed relative to their averages over all firms in the same province-industry-year cell. We find that SE are more productive and have larger firms. Compared to their non-SE counterparts, 1st and 2nd SE firms are 11% and 21% more productive, and, in terms of assets, 43% and 74% larger. Among the serial firms, 2nd-SE firms also have a higher TFP and more assets, revenue, and equity than 1st-SE firms.[28]

On average, the evidence is in line with the persistent-productivity view of Proposition 4 and not with the favored-entrepreneurs view of Proposition 5. The higher productivity of SE firms suggests that a positive correlation of TFP draws for entrepreneurs is quantitatively more important than any heterogeneity in the tightness of the financial constraint. Furthermore, the higher productivity of the 2nd-SE firms relative to the 1st-SE firms indicates high persistence ρ in the TFP draws, as required in Proposition 4. Thus, the evidence favors the Holmes and Schmitz (1995) view that entrepreneurs are serial for efficiency reasons: productivity is persistent across firms started by the same entrepreneur and better entrepreneurs start more firms.[29]

4.3.2 Concurrent versus Non-concurrent SE Firms

About half of the 2nd-SE firms are run concurrently with the 1st-SE firm, while in the other half of the cases the 1st-SE firm is closed when operating the 2nd-SE firm. In Table 7 we consider the TFP and equity of 1st-SE and 2nd-SE firms, depending on whether they are run concurrently or not, relative to non-SE firms. To this end, we run similar regressions on TFP and equity as in columns 1 and 4 of Table 6, albeit separately for concurrent and non-concurrent firms. We find that SE firms that are operated concurrently differ systematically from SE firms that are operated non-concurrently.

[28] In a contemporaneous paper Shaw and Sørensen (2022) document that similar properties hold for SEs in Denmark.

[29] These results are robust to restricting attention to firms started by 2009 instead of 2012, as in our benchmark sample. This sample change mitigates the potential problem associated with the fact that entrepreneurs who start a second firm after the end year of our data (2015) would be misclassified as non-serial (see footnote 8). The results are also robust to allowing more general controls for age. See Appendix Table A-6 for details.

Table 6: Performance of Firms, Registry and Inspection Data, 2008-2012.

	Sample with Inspection Data					Full Sample
	Log TFP	Log Assets	Log Revenue	Log Equity	Log Registered Capital	
	(1)	(2)	(3)	(4)	(5)	(6)
1st SE	0.11***	0.43***	0.35***	0.38***	0.37***	0.40***
2nd+ SE	0.21***	0.74***	0.58***	0.62***	0.60***	0.70***
Age	0.42***	0.19***	0.27***	0.14***	0.11***	0.07***
Age squared	-0.01***	-0.00***	-0.01***	-0.00***	-0.00***	-0.00***
Observations	12,669,977	12,669,977	12,669,977	12,669,977	12,669,977	36,213,534
Adjusted R-squared	0.03	0.10	0.09	0.08	0.07	0.05

Notes: Authors' calculations from the Registry and Inspection Data. The table compares the financial performance of 1st-SE and 2nd-SE firms, relative to non-SE firms. All variables, except age, are computed relative to their averages of all firms in the same province-industry-year cell. *** − statistically significant at the 1% level.

First, Table 7 shows that 1st-SE firms that are closed down when the 2nd-SE firm is started have a 43% lower TFP and 4% lower equity than those 1st-SE firms that are run concurrently with the 2nd-SE firm.[30] This finding is in line with the predictions of Proposition 6 that a relatively low productivity draw for the 1st-SE firm, as well as less equity, makes it more likely for the firm to be run non-concurrently with the 2nd-SE firm. Both concurrent and non-concurrent SE have on average higher TFP than non-serial entrepreneurs. However, the 1st SE firm of non-concurrent entrepreneurs is significantly less productive than the non-serial firms. This is in principle consistent with the negative selection of these firms (cf. Proposition 6), although this selection would have to be very strong quantitatively. However, we cannot rule out the possibility that there exists some ex-ante heterogeneity across entrepreneurs manifested in the choice of being non-concurrent and in the low TFP of the 1st SE firm. We explore the role of ex-ante heterogeneity in the next section.

Second, 2nd-SE firms that are operated by entrepreneurs who closed down their first firm (1st-SE) have a 13% higher TFP and 29% lower equity than those 2nd-SE firms run concurrently with the 1st-SE firm. This finding is also in line with the predictions of Proposition 6 − the combination of a good productivity draw for the 2nd-SE firm and relatively little equity makes it more likely that the entrepreneur closes the 1st SE firm and becomes a non-concurrent SE. We interpret this as evidence that firms face financial constraints.

4.4 Learning and Industry Choices

How are productivity and industry choice related? We first document that the persistence of TFP shocks is larger for serial firms started in the same industry than for serial firms started

[30]For TFP, for example, the difference for non-concurrent 1st SEs is given by subtracting the estimate in column (2) from the estimate in column (1), i.e., -0.15 - 0.28 = -0.43.

Table 7: TFP and Equity of Concurrent vs Non-concurrent SE Firms, Inspection Data, 2008-2012.

	Log TFP		Log Equity	
	Non-concurrent SE	Concurrent SE	Non-concurrent SE	Concurrent SE
	(1)	(2)	(3)	(4)
1st-SE	-0.15***	0.28***	0.49***	0.53***
2nd-SE	0.33***	0.20***	0.46***	0.74***
Age	0.42***	0.40***	0.13***	0.13***
Age squared	-0.01***	-0.01***	-0.00***	-0.00***
Observations	9,630,504	10,323,019	9,630,504	10,323,019
Adjusted R-squared	0.03	0.03	0.06	0.08

Notes: Authors' calculations from the Inspection Data. The table compares the TFP and equity of concurrent and non-concurrent 1st-SE and 2nd-SE firms, relative to non-SE. TFP and equity are computed relative to the averages of all firms in the same province-industry-year cell. *** − statistically significant at the 1% level.

in different industries. Recall that this is the key assumption in Proposition 7. Altogether, we have observations on 300,817 serial entrepreneurs. For non-concurrent SEs, we use the TFP of the 1st-SE firm in its last year of operation and the first year that 2nd-SE firm reports. For concurrent SE firms, we use information on 1st- and 2nd-SE firms for all years they both report. Table 8 shows that the persistence in TFP shocks is two-times higher for serial firms in the same I-O industries than for serial firms that are in different I-O industries. This confirms that the assumption on the persistence of TFP draws, summarized in equation (5) and underlying Proposition 7, is supported in the data.[31]

Consider now the implications from Proposition 7 regarding the ranking of TFP for industry stayers versus switchers. Table 9 shows that these predictions are born out: the TFPs of both 1st- and 2nd-SE firms are significantly higher when both firms are in the same I-O industry than when they are in different industries. In particular, the 1st-SE and 2nd-SE firms of industry stayers have respectively 51% and 75% higher TFP than non-serial firms. On the other hand, the 1st-SE and 2nd-SE firms of industry switchers both have 7% lower TFP than non-serial firms.

Since serial entrepreneurs who switch industries are actually less productive than non-serial entrepreneurs, the entire productivity premium of serial entrepreneurs must be a product of those who locate the 2nd-SE firm in the same I-O industry. Why, then, are the switchers willing to open the second firm? One possibility is that these entrepreneurs are largely favored individuals with non-skill advantages in the form of preferential access to borrowing. Recall from Proposition 5 that easier access to credit lowers the entry threshold, lowers TFP, and (conditional on TFP) increases the capital stock. Moreover, according to the learning

[31]Estimating a version of Table 8 separately for concurrent and non-concurrent entrepreneurs shows the same results for both groups: a persistence of 0.30 for stayers and 0.14 for switchers.

Table 8: Persistence in TFPs for 1st- and 2nd-SE Firms, at Entrepreneur Level.

	Log 2nd-SE TFP	
	Stayers	Switchers
	(1)	(2)
Log 1st-SE TFP	0.30***	0.14***
age	0.34***	0.30***
Age difference	0.33***	0.29***
Observations	106,427	194,390
Adjusted R-squared	0.10	0.04

Notes: Authors' calculations from the Inspection Data. 1st- and 2nd-SE firms are identified from the Registry Data. The table reports the persistence in the TFP of 1st- and 2nd-SE firms. All observations are for SEs who operate the 1st- and 2nd-SE firms for at least one year from 2008 to 2012. TFP is computed relative to the average of all firms in the same province-industry-year cell. Stayers are SEs whose I-O industry code of the 1st-SE firm and 2nd-SE firm are same. Switchers are SEs whose I-O industry code of the 1st-SE firm and 2nd-SE firm are different. *** − statistically significant at the 1% level.

Table 9: Performance of Serial Entrepreneurs: Industry Switchers vs. Stayers.

	Log TFP	Log Assets	Log Revenue	Log Equity	Log Registered Capital
	(1)	(2)	(3)	(4)	(5)
1st-SE Stayers	0.51***	0.38***	0.40***	0.34***	0.31***
1st-SE Switchers	-0.07***	0.46***	0.33***	0.40***	0.40***
2nd-SE Stayers	0.75***	0.69***	0.67***	0.58***	0.54***
2nd-SE Switchers	-0.07***	0.77***	0.53***	0.64***	0.64***
Age	0.42***	0.19***	0.27***	0.14***	0.11***
Age squared	-0.01***	-0.00***	-0.01***	-0.00***	-0.00***
Observations	12,669,977	12,669,977	12,669,977	12,669,977	12,669,977
Adjusted R-squared	0.03	0.10	0.09	0.08	0.07

Notes: Authors' calculations from the Registry and Inspection Data. The table compares the financial performance of 1st-SE and 2nd-SE firms of stayers and switchers, relative to non-SE firms. Stayers are SEs whose the I-O industry of the 2nd-SE firm is same as that of the 1st-SE firm. Switchers are SE whose I-O industry code of the 1st-SE and the 2nd-SE firms are different. All variables, except age, are computed relative to their averages of all firms in the same province-industry-year cell. *** − statistically significant at the 1% level.

hypothesis (Proposition 7) entrepreneurs with a low initial TFP are more likely to switch industry. It follows that favored individuals should be overrepresented among switchers. This insight suggests that industry switching can be used as a proxy for favored individuals.[32] The preponderance of favored individuals among switchers will in turn be a force for lower TFP and larger capital stock among switchers. In particular, switchers will have more capital than stayers, relative to their TFP. If the advantage is sufficiently large, the switchers could

[32]We provide a more rigorous discussion of this in Appendix C.

end up having larger average capital than the stayers, even if their average TFP is lower.

The results in Table 9 are consistent with this interpretation. The estimates in columns (2), (4), and (5) show that serial entrepreneurs who locate firms in distant industries have more assets, more equity, and more registered capital than their more productive peers who locate SE firms in the same (two-digit) industry. For example, the estimates in column (2) show that switchers − serial entrepreneurs who locate firms in different industries − have at least 8% more capital in the 1st- and 2nd-SE firms relative to stayers (i.e., serial entrepreneurs who locate firms in the same industry). This might seem surprising given that switchers have substantially lower TFP than stayers and should − if all entrepreneurs face the same cost of capital − install substantially less capital. However, as discussed above, the presence of favored entrepreneurs can rationalize why switchers have more capital even though their TFP is lower.

The evidence in Table 9 raises a puzzle for our simple theory of industry choice and TFP for serial entrepreneurs, namely, the growth in TFP between the 1st-SE and 2nd-SE firm should be *larger* for switchers than for stayers.[33] A simple extension of our theory of favored entrepreneurs can rationalize the empirical observation that switchers have lower growth in TFP between the 1st-SE and the 2nd-SE firms than stayers. Assume that the advantage of the favored entrepreneurs in access to credit increases over time. This implies that their entry threshold is lower for the 2nd-SE firm than for the 1st-SE firm. Clearly, if this improvement in credit access is sufficiently large, the positive selection associated with entry for the 2nd-SE firm should be smaller. Namely, the entrepreneur will choose to enter for a larger set of bad realizations of ε. If this effect is sufficiently large, it can counteract the positive expected TFP growth between the 1st-SE and 2nd-SE firms for switchers.

5 Additional Motives for Industry Choice

The analysis so far focused on learning as the main mechanism for industry choice. We now explore several additional motives for the industry choice of the 2nd-SE firm − risk diversification and complementarities through input-output linkages. Even though the model

[33]The reason is as follows. Note first that as long as $\rho \leq 1$, both stayers and switchers experiences (weakly) regression toward the switching threshold $\bar{z}$ for the expected value $E\{z_{2,s'} \mid z_{1,s}\}$. Second, since an entrepreneur staying in the same industry has $z_{1s} \geq \bar{z}$, she has $z_{1s} \geq E\{z_{2,s} \mid z_{1,s}\}$. In contrast, a switcher has $z_{1s} < \bar{z}$, implying that $z_{1s} < E\{z_{2,s'} \mid z_{1,s}\}$. Recall that selection into entry for the 2nd-SE firm will imply that the realized 2nd-SE TFP will be larger than the expected one conditional on entry (i.e., $E\{z_{2,s'} \mid z_{1,s}, \text{entry}\} \geq E\{z_{2,s'} \mid z_{1,s}\}$.) However, if the interest rate advantage of the favored entrepreneurs is not too large, this positive selection effect will be larger for switchers because they have a lower expected TFP and, hence, will choose to not enter for a larger set of bad realizations of ε. TFP growth between 1st-SE and 2nd-SE would therefore be larger for switchers than for stayers.

of serial entrepreneurship developed in the previous sections does not incorporate these two margins, they are both empirically relevant in understanding the industry choices of serial entrepreneurs and the TFPs of their 1st- and 2nd-SE firms.

5.1 Diversification of Risk

We start by studying the role of risk diversification for the industry choice of the 2nd-SE firm. We illustrate this motive with the aid of a simple portfolio choice model. To simplify the exposition, we focus on entrepreneurs who started and operated a firm in industry s in the first period and who have decided to solicit TFP draws for potential firms in a new industry $s' \neq s$.

We make three changes to the model relative to the previous analysis. First, we assume that the entrepreneur has linear-quadratic preferences over final wealth W given by the net savings plus profits in the firms. It follows immediately that the objective function of the entrepreneur can be expressed as a function of the mean and variance of W, $E\{u(W)\} = a \cdot E(W) - \frac{b}{2} \cdot Var(W)$, where $a > 0$ and $b > 0$.[34] Second, we assume that the entrepreneur obtains one (idiosyncratic) draw $z_{s'}$ for every industry $s' \in S \backslash s$. The entrepreneur chooses which of these firms to operate after having observed all draws.

Third, to embed a meaningful portfolio diversification motive, we assume that output of a firm in industry s has a stochastic industry-specific return to capital δ_s in addition to the regular production. The total output of the firm is therefore given by the sum $y_s + \delta_s k_s$, where $y_s = z_s^{1-\eta} \left(k_s^{1-\alpha} n_s^{\alpha}\right)^{\eta}$ is the regular deterministic idiosyncratic production (cf. equation (1)) with factor inputs k_s and n_s. Let W denote the realized final wealth for an entrepreneur with firms in industries s and s' and equity e in the beginning of period 2. W is then given by

$$W = \Pi\left(z_s, z_{s'}, e\right) + \delta_s k_s + \delta_{s'} k_{s'},$$

where the function Π denotes the profits in the 2nd period of operating firms with TFP z_s and $z_{s'}$ and equity e. Thus, $\Pi\left(z_s, z_{s'}, e\right) = y_s + y_{s'} - (n_s + n_{s'})w - (k_s + k_{s'} - e)R - \nu(\mathbb{1}_1 + \mathbb{1}_2)$, where $\mathbb{1}_j$ is an indicator function taking the value 1 if and only if firm $j \in \{1, 2\}$ is in operation.

Note that given the vector of realizations of TFP for potential 2nd-period firms, the variable Π is deterministic because factor allocations are made after observing TFP. However, the realizations of the industry-specific returns δ_s and $\delta_{s'}$ are observed *a*fter having made the industry choice and the factor allocations. Therefore, W is stochastic when industry and

[34] We implicitly assume that the range for W is such that utility is increasing in W for all realizations of W.

factor choices are made.

We assume that the returns δ_j have the same univariate distribution for all industries, although the covariance with initial industry, $Cov(\delta_{s'}, \delta_s)$, can differ across industries. Without loss of generality, we normalize the expected realization of δ_j to zero for all industries.

For an entrepreneur who operates two firms concurrently, the expected utility is given by

$$E\{u(W)\} = a \cdot \Pi\left(z_s, z_{s'}, e\right) - 2b \cdot \left[Var(\delta_s) + Cov\left(\delta_s, \delta_{s'}\right)\right].$$

Note that the expected utility is falling in the covariance between the industry-specific returns to capital. We state the main predictions of this section in the following proposition (the proof is in the appendix).

Proposition 8. *Consider the problem for an entrepreneur who has an existing firm in industry s and a set of idiosyncratic draws for potential firms $\{z_{s'}\}_{s' \in S \setminus s}$. The probability that the entrepreneur chooses industry s' is falling in the covariance $Cov(\delta_{s'}, \delta_s)$. Moreover, conditional on choosing industry s', the average TFP of the 2nd-SE firm is increasing in $Cov(\delta_{s'}, \delta_s)$.*

Proof. The proof is in Appendix B.6. □

Empirical Evidence on Diversification. The diversification theory requires that we measure the returns on firms in each industry, and the joint distribution (the variance-covariance matrix) of these returns. We focus on firms that are run concurrently one year after the establishment of the 2nd-SE because the entrepreneur would achieve risk diversification only when both firms are in operation. We construct an empirical measure of the return on capital in industry i in period t as:

$$r_{i,t} = \frac{profits_{i,t}}{assets_{i,t}},$$

where profits and assets are from the Inspection Data over the 2010-2012 period.[35] We calculate average returns for each I-O industry and use the empirical realizations to estimate the covariance matrix of returns.

Table 10 regresses TFP on observables for concurrently run firms with the omitted category being SE firms in the same I-O industry. Columns (1) and (2) convey the same message as the one in column (1) in Table 9 − SE firms in distant industries have significantly lower TFP than SE firms in the same industry. The third column of Table 10 adds the covariance

[35] We drop the financial crisis period.

between returns in the industries of the 1st-SE and 2nd-SE firms as a control variable for 2nd-SE firm TFP. The table shows that the TFP of the 2nd-SE firm is increasing in the covariance. This result confirms that entrepreneurs require a risk premium for choosing industries with more positive correlation with the entrepreneur's initial industry. Note that the measure of covariance in the regression is standardized so that this variable has mean zero and a standard deviation equal to one. The coefficient of 0.31 therefore implies that increasing this variable (the covariance) by one standard deviation (of the cross-sectional dispersion in covariance) will increase TFP by 31%. This effect is significant, both from an economic and a statistical point of view. This result is robust to including all SE firms in the regression, including firms that are not run concurrently.

Table 10: Risk Diversification and TFP for 1st- and 2nd-SE Firms, at Entrepreneur Level.

	log 1st-SE TFP (1)	log 2nd-SE TFP (2)	log 2nd-SE TFP (3)
Switchers	-0.47***	-0.67***	-0.64***
Switchers×Covariance	–	–	0.31***
Age	0.08***	0.72***	0.72***
Age squared	-0.00***	-0.03***	-0.03***
Observations	262,993	262,993	262,993
Adjusted R-squared	0.00	0.03	0.03

Notes: Authors' calculations from the Inspection Data. 1st- and 2nd-SE firms are identified from the Registry Data. All observations are SE who operate the 1st- and 2nd-SE firms concurrently for at least one year from 2008 to 2012. Switchers are SE whose I-O industry code of the 1st-SE and the 2nd-SE are different. TFP is computed relative to the averages of all firms in the same province-industry-year cell. The variable Covariance is the covariance of the return of assets between each two industries. This variable is standardized to have mean zero and a standard deviation of one. *** – statistically significant at the 1% level.

5.2 Upstream-Downstream Integration and Input-Output Complementarities

A natural extension of our simple multi-industry model in Section 5.1 is to allow for input-output linkages and trade in intermediate goods. Consider two firms that have large potential gains from trade with each other. Rather than modeling such trade- and input-output linkages explicitly, we simply appeal to Williamson (1975) transaction-cost theory and postulate that having joint ownership of these firms can mitigate the potential information asymmetries and transaction costs of trade.[36] This implies that an entrepreneur with a firm in

[36]This might be considered as a hybrid approach to mitigating transaction costs, in between pure trade and full vertical integration.

industry s has a comparative advantage of operating in industries that trade with firms in industry s.

If the only difference across potential industries s' is the strength of their linkages with industry s, then industries with stronger links will be chosen more often. Note, however, that it is difficult to obtain sharp predictions for TFP of the 2nd-SE firm because the theory is silent about which firm benefits from the linkages, be it the 1st-SE, 2nd-SE, or both.

Measuring upstream and downstream integration. We measure the downstream and upstream industry linkages using the methodology in Fan and Lang (2000). Taking a serial entrepreneur with a 1st-SE firm in industry i and a 2nd-SE firm in industry j, the "upstream" index is defined as the dollar value of industry j's output required to produce 1 dollar's worth of industry i's output while the "downstream" index is defined as the dollar value of industry i's output required to produce 1 dollar's worth of industry j's output. We use the 2007 Chinese Input-Output tables to compute these indices.

Measuring input and output complementarities. In order to study any potential input and output complementarity links between the firms of serial entrepreneurs, we construct the following two indices. The "output complementarity" index is the correlation coefficient between b_{ik} and b_{jk}, where b_{ik} (b_{ik}) is the percentage of industry i (j) output supplied to each intermediate industry k. This index captures the degree to which industries i and j share outputs. The "input complementarity" index on the other hand is defined as the correlation coefficient between v_{ik} and v_{jk}, where v_{ik} (v_{ik}) is the percentage of inputs from each intermediate industry k used in industry i (j) output. This index captures the degree to which industries i and j share inputs. We use the 2007 Chinese Input-Output tables to compute these indices.

Measuring the excess probability of industry choice. In order to study the quantitative importance of the measures described above in determining the industry choice of a 2nd-SE firm, we construct an excess probability measure: the normalized probability of starting the 2nd-SE firm in industry j, given that the 1st-SE firm is in industry i. Consider serial entrepreneurs with a 1st-SE firm in industry i and a 2nd-SE firm in industry j. We calculate the percentage of serial entrepreneurs that move from i to j each year from 1995-2015 when they start their 2nd-SE firm − this is computed as the number of serial entrepreneurs from i to j divided by total number of serial entrepreneurs in industry i. This measure is then normalized by the share of industry j for non-SE entrants in that year.

Empirical Evidence on the Role of Linkages. We close this section by studying how industry linkages and diversification motives shape the probability of industry choice for the 2nd-SE firm conditional on the location of the 1st-SE. Table 11 reports the results of a regression of the excess probability measure on our measure of covariance, downstream and upstream integration, and complementarity.[37] In the regression, we control for the industry of the 1st-SE firm and the year the 2nd-SE firm is established. The results reported in column (5) indicate that serial entrepreneurs are more likely to start their 2nd-SE firm in an industry that is upstream integrated, downstream integrated, and complementary with the industry of their 1st-SE firm. These results confirm that entrepreneurs should be more likely to choose integrated industries. Moreover, Table 11 confirms the prediction of Proposition 8 that switchers should be more likely to choose industries that are negatively correlated with the initial industry (see the Covariance row, where this variable is defined as in Table 10).

Table 11: Industry Choice: Diversification, and Business Linkages, 1995-2015.

	Excess Probability				
	(1)	(2)	(3)	(4)	(5)
Downstream Integrated	0.86***	–	–	–	0.78***
Upstream Integrated	–	0.63***	–	–	0.51***
Complementarity	–	–	0.52***	–	0.41***
Covariance	–	–	–	-0.06	-0.08*
Observations	348,171	348,171	348,171	348,171	348,171
Adjusted R-squared	0.02	0.01	0.02	0.01	0.03

Note: We control for the industry of the 1st-SE firm and the year the 2nd-SE firm is established. The weight for regressions is the number of new entrants of each industry of the 1st-SE.

6 Conclusion

This paper uses data on the universe of all firms in China to document key facts about entrepreneurship and serial entrepreneurship in China since the early 1990s. We examine these data through the lens of a model of serial entrepreneurship in which potential entrepreneurs face capital market frictions in the form of collateral constraints. We highlight two key factors potentially driving serial entrepreneurship: persistence in entrepreneurial skills and learning, and preferential access to finance or other scarce inputs. These drivers have sharply different implications for the outcomes of serial entrepreneurs relative to their non-serial counterparts.

[37] Input and output complementarity are combined into one measure.

We find that serial entrepreneurs in China are better than non-serial entrepreneurs. We interpret this as evidence that persistence and learning create positive selection of serial entrepreneurs. Serial entrepreneurs who start their firms in the same two-digit industry enjoy fifty percent higher productivity than non-serial firms in the same industries. In principle, serial entrepreneurs who switch industries should also be better, albeit not as good as those who remain in the same industry. We find that the productivity of switchers is lower than stayers, but surprisingly, significantly lower than non-serial firms. While frictions upstream and downstream may be contributing to the lower productivity of switchers, the most likely explanation for the negative selection of these entrepreneurs into serial entrepreneurship is preferential access to finance.

How do we explain these diametrically opposite forces shaping entrepreneurial behavior? In this paper, we assume that frictions are the same across space as well as industries. They may, in fact, differ, which has implications for selection into entrepreneurship (Brandt, Kambourov and Storesletten (forthcoming)), and thus the properties of serial entrepreneurs. In future work, we plan to examine these differences.

Finally, in this paper we analyze serial entrepreneurship through the lens of entrepreneurs who establish second firms as individual investors. Firm owners can also start second firms with investments through the enterprises they control. In other contexts (Bena and Ortiz-Molina (2013)), these differences were key to explaining differences between new firms.

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Online Appendix for

"Serial Entrepreneurship in China"

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A Tables and Figures

Table A-1: Number of Firms in the Inspection Data, 2008-2012.

Year	Number of Firms
2008	1711562
2009	2002896
2010	2476916
2011	3064303
2012	3414300
Total	12669977

Notes: Authors' calculations from the Inspection Data.

Table A-2: Reporting Ratio of Inspection Data, Different Type of Entrepreneur.

Year	Non-SE	1st-SE	2nd-SE
2008	43.78%	46.15%	43.61%
2009	45.73%	48.01%	45.25%
2010	49.63%	51.66%	47.82%
2011	53.65%	55.98%	50.30%
2012	53.58%	55.74%	49.57%

Notes: Authors' calculations from the Registry Data and the Inspection Data.

Table A-3: Share of Registered Capital, by Ownership Type, 1995-2015.

Year	Total (Trillion)	Unregistered(%)	Individual(%)		Enterprise(%)		Share of baseline sample: (2)+(3)
		(1)	Single (2)	Multiple (3)	Single (4)	Multiple (5)	
1995	13.50	50.45	2.95	7.99	23.73	14.88	10.94
1996	14.87	50.17	3.13	9.50	21.74	15.46	12.63
1997	16.31	48.51	3.34	10.96	21.19	15.99	14.30
1998	19.03	43.33	3.35	18.55	19.18	15.59	21.90
1999	21.34	42.73	3.43	18.89	18.94	16.00	22.32
2000	24.63	39.45	3.36	18.82	18.60	19.77	22.18
2001	26.41	38.76	3.82	17.78	19.05	20.60	21.60
2002	28.75	37.85	4.63	19.16	16.50	21.86	23.78
2003	34.14	34.98	4.64	22.12	17.78	20.48	26.76
2004	35.98	35.09	5.21	22.55	18.01	19.13	27.76
2005	39.32	34.12	5.45	23.95	17.46	19.03	29.40
2006	43.26	33.30	5.76	24.67	16.94	19.34	30.43
2007	48.77	34.45	5.56	23.56	16.69	19.73	29.12
2008	52.59	33.77	5.81	23.40	16.72	20.30	29.21
2009	58.73	31.95	5.99	24.68	16.80	20.59	30.66
2010	67.79	29.55	6.14	26.49	16.62	21.20	32.62
2011	78.07	27.33	6.16	28.27	16.34	21.91	34.43
2012	87.60	25.72	6.21	29.15	16.35	22.57	35.36
2013	99.82	23.84	6.46	30.40	15.88	23.42	36.86
2014	125.69	20.67	7.77	33.57	14.89	23.10	41.34
2015	161.53	18.39	9.26	35.06	13.94	23.35	44.32

Notes: Authors' calculations from the Registry Data.

Table A-4: Entry and Exit of Firms, by Serial Entrepreneur Status, Registry Data, 1995-2015.

Year	Non-SE					1st-SE					2nd-SE				
	Survival	New	Exit	Entry rate (%)	Exit rate (%)	Survival	New	Exit	Entry rate (%)	Exit rate (%)	Survival	New	Exit	Entry rate (%)	Exit rate (%)
1995	255281	79230	1449	44.64	0.82	101392	32720	378	47.39	0.55	13649	4298	47	45.73	0.50
1996	342187	90733	3827	35.54	1.50	138605	38289	1076	37.76	1.06	19507	6064	206	44.43	1.51
1997	445856	111720	8051	32.65	2.35	183287	46966	2284	33.88	1.65	28164	9047	390	46.38	2.00
1998	591741	160711	14826	36.05	3.33	245842	67187	4632	36.66	2.53	42501	15239	902	54.11	3.20
1999	744789	180349	27301	30.48	4.61	310848	74474	9468	30.29	3.85	61995	21272	1778	50.05	4.18
2000	941225	234011	37575	31.42	5.05	391935	93894	12807	30.21	4.12	90756	31885	3124	51.43	5.04
2001	1177521	291674	55378	30.99	5.88	485266	111939	18608	28.56	4.75	130942	46014	5828	50.70	6.42
2002	1466502	360214	71233	30.59	6.05	587600	128996	26662	26.58	5.49	187014	64731	8659	49.43	6.61
2003	1826715	453275	93062	30.91	6.35	706033	154942	36509	26.37	6.21	268039	93320	12295	49.90	6.57
2004	2248106	538832	117441	29.50	6.43	835496	173297	43834	24.55	6.21	365302	114803	17540	42.83	6.54
2005	2661806	558619	144919	24.85	6.45	949500	167150	53146	20.01	6.36	475850	135104	24556	36.98	6.72
2006	3082195	599348	178959	22.52	6.72	1047088	159037	61449	16.75	6.47	591328	148364	32886	31.18	6.91
2007	3404725	599751	277221	19.46	8.99	1109610	150698	88176	14.39	8.42	691569	151033	50792	25.54	8.59
2008	3775775	647212	276162	19.01	8.11	1177517	152671	84764	13.76	7.64	795829	159424	55164	23.05	7.98
2009	4343514	838904	271165	22.22	7.18	1277974	181356	80899	15.40	6.87	948327	210248	57750	26.42	7.26
2010	5073756	1009311	279069	23.24	6.42	1387115	188143	79002	14.72	6.18	1155879	268499	60947	28.31	6.43
2011	5932872	1175431	316315	23.17	6.23	1487933	184922	84104	13.33	6.06	1393257	308286	70908	26.67	6.13
2012	6747935	1198632	383569	20.20	6.47	1554932	160251	93252	10.77	6.27	1620835	316159	88581	22.69	6.36
2013	8093650	1707083	361368	25.30	5.36	1645899	175307	84340	11.27	5.42	1946573	418230	92492	25.80	5.71
2014	10233045	2433003	293608	30.06	3.63	1756998	173709	62610	10.55	3.80	2536705	664817	74685	34.15	3.84
2015	12605107	2703599	331537	26.42	3.24	1783434	85933	59497	4.89	3.39	3210127	761557	88135	30.02	3.47

Notes: Authors' calculations from the Registry Data. Survival measures the number of firms in a given year while New and Exit denote the number of new and exiting firms in a given year, respectively.

Table A-5: Debt-Equity Ratio, Capital, and Relative TFP, Conditional on Equity, Inspection Data, 2008-2012.

	Log Assets	Debt-Equity Ratio
	(1)	(2)
Log TFP	0.04***	0.18***
2nd quarter of equity	1.27***	-2.00***
3rd quarter of equity	1.87***	-2.10***
4th quarter of equity	3.39***	-2.63***
TFP × 2nd quarter of equity	0.00***	-0.08***
TFP × 3rd quarter of equity	0.00***	-0.09***
TFP × 4th quarter of equity	0.01***	-0.13***
Age	0.07***	0.14***
Age square	-0.00***	-0.00***
Observations	12,669,977	12,669,926
Adjusted R-squared	0.52	0.00

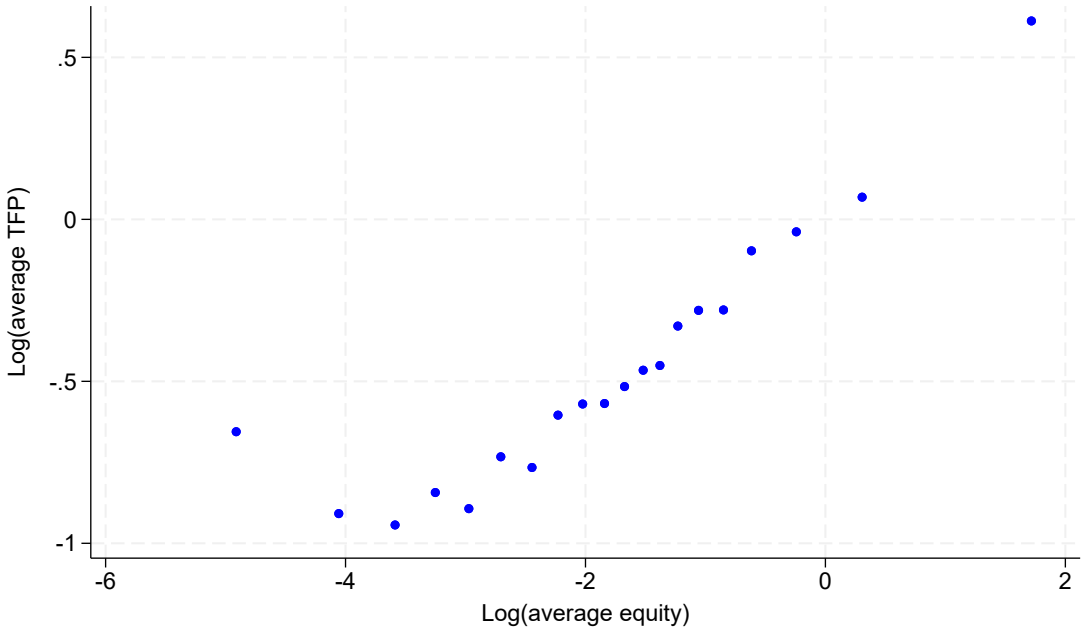
Notes: Authors' calculations from the Inspection Data. The table reports the relationship between assets and the debt-equity ratio and TFP. The results are computed for different quarters in the equity distribution. All variables, except age, are computed relative to their averages of all firms in the same province-industry-year cell. *** – statistically significant at the 1% level.

Table A-6: Performance of Serial Entrepreneurs: Robustness Test for Age Effects

	Log TFP			
	benchmark	age quartic	age dummy	firms established 2009 or earlier
	(1)	(2)	(3)	(4)
1st-SE	0.11***	0.11***	0.11***	0.13***
2nd-SE	0.21***	0.17***	0.17***	0.23***
Age	0.42***	1.00***	–	0.28***
Age square	-0.01***	-0.09***	–	-0.01***
Age cubic	–	0.00***	–	–
Age quartic	–	-0.00***	–	–
Age fixed effects	No	No	Yes	No
Observations	12,669,977	12,669,977	12,669,976	10,054,120
Adjusted R-squared	0.03	0.04	0.04	0.01

Notes: Authors' calculations from the Registry and Inspection Data. The table compares the performance of 1st-SE and 2nd-SE firms, relative to non-SE firms. All variables, except age, are computed relative to their averages of all firms in the same province-industry-year cell. Column (1) is same to the Table 6. Column (2) adds age cubic and quartic as additional controls. Column (3) adds age fixed effects. Column (4) includes firms that are established in 2009 or earlier. *** – statistically significant at the 1% level.

Figure A-1: Equity and TFP, Newly Established Non-Serial and 1st-SE Firms.



Notes: The figure reports a bin scatter plot of the log average TFP and log average equity for non-serial and 1st-SE firms that are less than four years old in the Inspection Data. Based on their equity, firms are divided into 20 ventiles, and the figure reports the averages for each ventile. Data source: Inspection Data.

B Theoretical Results

B.1 Proof of Proposition 1

Consider first an unconstrained entrepreneur. Assuming that $k < \lambda e$, the entrepreneur's problem is

$$\begin{aligned}\Pi(e,z;1) &= \max_{k,n}\{y - wn - Rk\} + eR \\ &= \max_{k,n}\left\{z^{1-\eta}\left(k^{1-\alpha}n^{\alpha}\right)^{\eta} - wn - R(k-e)\right\}.\end{aligned}$$

The first-order conditions are given by

$$\alpha\eta y = wn, \tag{B-1}$$

$$(1-\alpha)\eta y = Rk, \tag{B-2}$$

Plugging this back into the production function yields an expression for output in terms of z and equity. The optimal allocations follow directly from eq. (B-1)-(B-2) and are given by equation (B-3),

$$\begin{aligned}y^*(e,z,1) &= z\cdot\left(\frac{(1-\alpha)\eta}{R}\right)^{\frac{(1-\alpha)\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\eta}} \\ k^*(e,z,1) &= z\cdot\left(\frac{(1-\alpha)\eta}{R}\right)^{\frac{1-\alpha\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\eta}} \equiv zk^* \\ n^*(e,z,1) &= z\cdot\left(\frac{(1-\alpha)\eta}{R}\right)^{\frac{(1-\alpha)\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{\frac{1-\eta(1-\alpha)}{1-\eta}}\end{aligned} \tag{B-3}$$

and where profits are

$$\Pi(e,z,1) = z\cdot(1-\eta)\cdot\left(\frac{(1-\alpha)\eta}{R}\right)^{\frac{(1-\alpha)\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\eta}} + Re$$

Consider now the unconstrained entrepreneur's decision whether or not to enter. The entrepreneur will enter if profits exceed the opportunity cost, which is depositing the equity in the bank. Given the prices and state variables, the condition $\Pi(z,e,1) - \nu \geq Re$ implies a cutoff z^* such that all potential entrepreneurs with $z \geq z^*$ will choose to operate firms, where z^* is given by

$$z^* \equiv \frac{v}{1-\eta}\left(\frac{(1-\alpha)\eta}{R}\right)^{-\frac{(1-\alpha)\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{-\frac{\alpha\eta}{1-\eta}} = \frac{\eta}{1-\eta}\frac{1-\alpha}{R}\frac{\nu}{k^*}$$

This threshold is independent of equity since equity is irrelevant for the unconstrained entrepreneur. Moreover, the threshold is increasing in the wage rate (since higher wages lower profits) and increasing in R (since higher returns on deposits increase the alternative value of equity). The unconstrained entrepreneur will install a capital stock given by $k^*(e,z,1) = zk^*$ (see equation (B-3)). It follows that the potential entrepreneur will be an unconstrained entrepreneur and operate the firm if and only if two conditions are simultaneously satisfied: (1) $z \geq z^*$ and (2) $\lambda e \geq z\cdot k^*$. Namely, both TFP and equity must be sufficiently large. Moreover, it follows that the lower bound for equity for an unconstrained entrepreneur is $\underline{e} \equiv z^*k^*/\lambda = \nu(1-\alpha)\eta/[(1-\eta)\lambda R]$.

Next, consider a constrained entrepreneur who is constrained in terms of borrowing, i.e., $k = \lambda e$ and $b = (\lambda - 1)\, e$. This entrepreneur solves the problem

$$\Pi\left(e, z; 0\right) = \max_{n} \left\{ z^{1-\eta} \left(\left(\lambda e\right)^{1-\alpha} n^{\alpha} \right)^{\eta} - wn - R\left(\lambda - 1\right) e \right\}.$$

The first-order condition for employment n, equation (B-1), applies, while equation (B-2) becomes an inequality, $R\lambda e < (1-\alpha)\,\eta y$. For constrained entrepreneurs the optimal allocations are given by equation (B-4),

$$\begin{aligned} y_c^* &= z^{\frac{1-\eta}{1-\alpha\eta}} \left(\lambda e\right)^{\frac{(1-\alpha)\eta}{1-\alpha\eta}} \left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\alpha\eta}} \\ k_c^* &= \lambda e \\ n_c^* &= z^{\frac{1-\eta}{1-\alpha\eta}} \left(\lambda e\right)^{\frac{(1-\alpha)\eta}{1-\alpha\eta}} \left(\frac{\alpha\eta}{w}\right)^{\frac{1}{1-\alpha\eta}} \\ \Pi\left(e, z; 0\right) &= \left(1-\alpha\eta\right) z^{\frac{1-\eta}{1-\alpha\eta}} \left(\lambda e\right)^{\frac{(1-\alpha)\eta}{1-\alpha\eta}} \left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\alpha\eta}} - R\left(\lambda - 1\right) e, \end{aligned} \tag{B-4}$$

where the subscript c denotes "constrained."

The analysis of the unconstrained and constrained cases implies that the potential entrepreneur will be constrained if and only if

$$\lambda e < z k^*.$$

Note that the return to equity for constrained entrepreneurs exceeds R.

Consider now the entry decision for the constrained entrepreneurs. The entrepreneur will enter if operating the firm is better than depositing the equity, i.e., if $\Pi\left(e, z, 0\right) - \nu \geq Re$. This condition implies a threshold function $z^*\left(e\right)$ given by

$$z^*\left(e\right) \geq \left(\frac{\nu + R\lambda e}{1-\alpha\eta}\right)^{\frac{1-\alpha\eta}{1-\eta}} \left(\lambda e\right)^{-\frac{(1-\alpha)\eta}{1-\eta}} \left(\frac{w}{\alpha\eta}\right)^{\frac{\alpha\eta}{1-\eta}}.$$

Equity and better financial markets (larger λ) affects the threshold for constrained entrepreneurs in two opposing ways. On the one hand, a larger equity and/or a larger λ increase the value of the firm. This tends to reduce the threshold. On the other hand, a larger equity and/or a larger λ increase the opportunity cost of deposits, which tends to decrease the threshold. The former effect dominates and the comparative statics of the threshold with respect to e is given by

$$\frac{\partial \ln\left(z^*\left(e\right)\right)}{\partial \ln e} = \frac{1-\alpha\eta}{1-\eta} \frac{R\lambda e}{\nu + R\lambda e} - \frac{\left(1-\alpha\right)\eta}{1-\eta} \leq 0,$$

where the inequality is strict for $e < e^* = \frac{\nu}{1-\eta} \frac{(1-\alpha)\eta}{R}$ and holds with equality for $e = e^*$.

B.2 Proof of Proposition 2

The proof that SE firms have higher TFP than non-SE firms (part 1 of Proposition 2) relies on the following lemma for conditional expectations.

Lemma 1. *If $\rho \geq 0$ then*

$$E\{z | z \geq a \ \textit{and} \ G(z) + \epsilon \geq b\} \geq E\{z | z \geq a\},$$

where G is a monotone increasing function, z and ε are stochastic variables, and a and b are constants.

When setting $G(z) = \rho z$ and $a = b = z^*$ and interpreting z and $\rho z + \epsilon$ as the TFP draw in the first and second period, respectively, Lemma 1 implies the key result of Part 1 that 1st-SE firms will on average have higher TFP than non-serial firms provided that $\rho > 0$. A similar argument establishes also that 2nd-SE firms are on average more productive than non-serial firms when $\rho > 0$. To see this, note that Lemma 1 also implies that $E[z_2|z_2 \geq b, z_2 \geq a - \epsilon/\rho] \geq E[z_2|z_2 \geq b]$, where $\tilde{\varepsilon} \equiv -\varepsilon/\rho$ is a stochastic variable and $z_1 = z_2/\rho - \varepsilon/\rho = z_2/\rho + \tilde{\varepsilon}$.

- **Proof of Lemma 1.** Proving Lemma 1 amounts to proving that if $g(z)$ is monotone increasing in z then

$$E[z|z \geq a, G(z) + \epsilon \geq b] \geq E[z|z \geq a],$$

where Z is a stochastic variable and a and b are constants.

The main idea is to show that (1) adding the condition $G(z) + \epsilon \geq b$ is equivalent to multiplying an increasing function $h(z)$ to the pdf conditional on $z \geq a$, denoted as $f(z)$ and (2) generally, if we multiply pdf $f(z)$ by an increasing function $h(z)$ to get a new pdf $g(z)$, then $g(z)$ first order dominates $f(z)$ and leads to higher expected z.

First, denote the unconditional pdf and cdf of z as $i(z)$ and $I(z)$, and the pdf and cdf of ϵ as $j(z)$ and $J(z)$. The pdf conditional on $z \geq a$ can then be expressed as

$$f(z) = \frac{i(z)}{1 - I(a)}, z \geq a.$$

Then the pdf of z conditional on $z \geq a$ and $G(z) + \epsilon \geq b$ is

$$\begin{aligned} g(z) &= f(z) \frac{\int_{b-G(z)}^{\infty} j(\epsilon) d\epsilon}{\int_a^{\infty} f(x) \int_{b-G(x)}^{\infty} j(\epsilon) d\epsilon dx} = f(z) \frac{1 - J(b - G(z))}{\int_a^{\infty} f(x)(1 - J(b - G(x))) dx} \\ &\doteq f(z) \frac{h(z)}{\int_a^{\infty} f(x) h(x) dx}, \end{aligned}$$

where $h(z) = 1 - J(b - G(z))$ is an increasing function of z because $G(z)$ is increasing in z.

Next, we illustrate the impacts of multiplying $h(z)$ to a pdf $f(z)$. Define

$$g(z) = \frac{f(z) h(z)}{\int f(x) h(x) d(x)} \doteq \frac{f(z) h(z)}{H},$$

where H is a constant to turn $\int g(z) dz = 1$ and make g also a pdf.

Third, we show that g first order dominate (FOD) f, i.e., for any z, we have $G(z) < F(z)$. If z is small, such that $h(z) \leq H$, then

$$G(z) = \int^z \frac{f(x) h(x)}{H} dx < \int^z \frac{f(x) h(z)}{H} dx = F(z).$$

If z is large, such that $h(z) > H$, then

$$1 - G(z) = \int_z \frac{f(x) h(x)}{H} dx > \int_z \frac{f(x) h(z)}{H} dx = 1 - F(z).$$

FOD implies higher expected value. To see this, note that for any z and $F(z)$, we can find a corresponding $y > z$ such that $G(y) = F(z)$, because $G(z) < F(z)$ and G is increasing. Then

$$E[z|F] = \int z dF(z) = \int z dG(y) < \int y dG(y) = \int z dG(z) = E[z|G].$$

QED

B.3 Optimal allocations under financial frictions

Proof of Proposition 3. The maintained assumption is that the entrepreneur entered and operated firm 1 in period 1. If $z_2 > z_1$, it will be strictly more profitable to operate firm 2 than firm 1. The entrepreneur will therefore enter and operate firm 2 regardless whether or not firm 1 is operated. It follows that $Z(z_1, e) \leq z_1$. Propostion 1 implies that firm 2 would not be operated if $z_2 < z^*$. From now on we focus on the case when $z^* \leq z_2 \leq z_1$.

Suppose first that $\lambda e \geq (z_2 + z_1) k^*$. Propostion 1 implies that it is better to operate each firm with capital $z_2 k^*$ and $z_1 k^*$, respectively, than depositing the equity earning rate R. Since equity is sufficient to fund both firms, this allocation is also feasible. This lower bound on z_2 is independent of e and z_1.

Suppose now that $\lambda e < (z_2 + z_1) k^*$. Propostion 1 then implies that if the entrepreneur is operating both firms then she will be constrained: $b = (\lambda - 1) e$. The optimal employment would be to allocate capital and labor so as to equate the marginal product of labor in each firm to the wage rate. This implies that for each firm j,

$$n_j = (z_j)^{\frac{1-\eta}{1-\alpha\eta}} (k_j)^{\frac{(1-\alpha)\eta}{1-\alpha\eta}} \left(\frac{\alpha\eta}{w}\right)^{\frac{1}{1-\alpha\eta}}.$$

Moreover, the entrepreneur's equity would be distributed across the firms so as to equalize the marginal product of capital across firms. This implies

$$(1-\alpha)\eta (z_2)^{1-\eta} \frac{\left(k_2^{1-\alpha} n_2^{\alpha}\right)^{\eta}}{k_2} = (1-\alpha)\eta (z_1)^{1-\eta} \frac{\left(k_1^{1-\alpha} n_1^{\alpha}\right)^{\eta}}{k_1}$$

which in turn implies $k_2 = \frac{z_2}{z_1} k_1$. Since we hypothesize $k_1 + k_2 = \lambda e$, it follows that $\lambda e = \left(\frac{z_2}{z_1} + 1\right) k_1$, implying

$$k_2 = \frac{z_2}{z_2 + z_1} \lambda e \qquad \text{and} \qquad k_1 = \frac{z_1}{z_2 + z_1} \lambda e.$$

Maintaining that $\lambda e \leq (z_2 + z_1) k^*$, we now consider two cases.

Suppose first that equity is sufficiently large that the entrepreneur is unconstrained when operating one firm, i.e., $\lambda e \geq z_1 k^*$ so $\lambda e \in [z_1 k^*, (z_2 + z_1) k^*)$. The entrepreneur would then operate two firms if and only if

$$\Pi\left(\frac{z_2}{z_2 + z_1} \lambda e, z_2; 0\right) + \Pi\left(\frac{z_1}{z_2 + z_1} \lambda e, z_1; 0\right) - 2\nu \geq \Pi(z_1; 1) + Re - \nu, \tag{B-5}$$

where the function $\Pi(k, z; 0)$ denotes profits net of the operating cost ν from a constrained entrepreneur with equity e operating a firm with capital $k = \lambda e$ and TFP z,

$$\Pi(\lambda e, z, 0) = (1 - \alpha\eta) z^{\frac{1-\eta}{1-\alpha\eta}} (\lambda e)^{\frac{(1-\alpha)\eta}{1-\alpha\eta}} \left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\alpha\eta}} - R(\lambda - 1) e.$$

Moreover, the function $\Pi(z; 1)$ denotes profits net of the operating cost ν from an unconstrained entrepreneur

operating a firm with TFP z,

$$\Pi(z;1) \equiv z \cdot (1-\eta) \cdot \left(\frac{(1-\alpha)\eta}{R}\right)^{\frac{(1-\alpha)\eta}{1-\eta}} \left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\eta}}.$$

Simple algebra establishes that the condition (B-5) is equivalent to the following lower bound on z_2,

$$\begin{aligned} z_2 &\geq Z(z_1,e) \equiv \left(z_1 \cdot \frac{1-\eta}{1-\alpha\eta}\left(\frac{(1-\alpha)\eta}{R}\right)^{\frac{(1-\alpha)\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\eta}} + \frac{R\lambda e+\nu}{1-\alpha\eta}\right)^{\frac{1-\alpha\eta}{1-\eta}} (\lambda e)^{-\frac{(1-\alpha)\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{-\frac{\alpha\eta}{1-\eta}} \quad \text{(B-6)} \\ &= \left(1+\frac{1-\eta}{1-\alpha\eta}\left(\frac{zk^*}{\lambda e}-1\right)+\frac{(1-\alpha)\eta}{1-\alpha\eta}\frac{1}{\lambda e}\frac{v}{R}\right)^{\frac{1-\alpha\eta}{1-\eta}}\frac{\lambda e}{k^*}-z \end{aligned}$$

Simple algebra establishes that

$$\frac{\partial Z(z,e)}{\partial z} = \left(1+\frac{1-\eta}{1-\alpha\eta}\left(\frac{zk^*}{\lambda e}-1\right)+\frac{1-\eta}{1-\alpha\eta}\frac{z^*k^*}{\lambda e}\right)^{\frac{1-\alpha\eta}{1-\eta}-1} - 1 > 0,$$

which is positive given the maintained assumption that $\lambda e \leq z_1 k^*$. This implies that the lower bound $Z(z_1,e)$ is increasing in z_1 whenever $\lambda e \in [z_1 k^*, (z_2+z_1)k^*)$.

In terms of equity, simple algebra establishes that

$$k^* = \left(\frac{(1-\alpha)\eta}{R}\right)^{\frac{1-\alpha\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{\frac{\alpha\eta}{1-\eta}}$$

$$\begin{aligned} \frac{\partial Z}{\partial e} &= -\left(z_1k^* - \lambda e + \frac{(1-\alpha)\eta}{1-\eta}\frac{\nu}{R}\right) \\ &\quad \cdot\frac{R\lambda}{1-\alpha\eta}\left(\frac{1-\eta}{1-\alpha\eta}\frac{Rk^*}{(1-\alpha)\eta}z_1+\frac{R\lambda e+\nu}{1-\alpha\eta}\right)^{\frac{1-\alpha\eta}{1-\eta}-1}(\lambda e)^{-\frac{1-\alpha\eta}{1-\eta}}\left(\frac{\alpha\eta}{w}\right)^{-\frac{\alpha\eta}{1-\eta}} \\ &< 0, \end{aligned}$$

where the inequality follows from the maintained assumption that $z_1k^* \geq \lambda e$. This implies that the lower bound $Z(z_1,e)$ is falling in e whenever $\lambda e \in [z_1k^*, (z_2+z_1)k^*)$.

Finally, consider the case when equity is sufficiently small that the entrepreneur would be constrained even when operating one firm, i.e., $\lambda e < z_1 k^*$. In this case, the entrepreneur would operate two firms if

$$\Pi\left(\frac{z_2}{z_2+z_1}\lambda e, z_1; 0\right) + \Pi\left(\frac{z_1}{z_2+z_1}e, z_1; 0\right) - 2\nu \geq \Pi(e, z_1; 0) - \nu. \quad \text{(B-7)}$$

Standard algebra establishes that the condition (B-7) is equivalent to the lower bound $z_2 \geq Z(z_1,e)$, where

$$Z(z_1,e) \equiv \left((z_1)^{\frac{1-\eta}{1-\alpha\eta}} + \left(\frac{w}{\alpha\eta}\right)^{\frac{\alpha\eta}{1-\alpha\eta}}\frac{\nu}{1-\alpha\eta}(\lambda e)^{-\frac{(1-\alpha)\eta}{1-\alpha\eta}}\right)^{\frac{1-\alpha\eta}{1-\eta}} - z_1. \quad \text{(B-8)}$$

It is immediate that $Z(z_1,e)$ is monotone increasing in z_1 and monotone falling in e in this range. This completes the proof of Proposition 3.

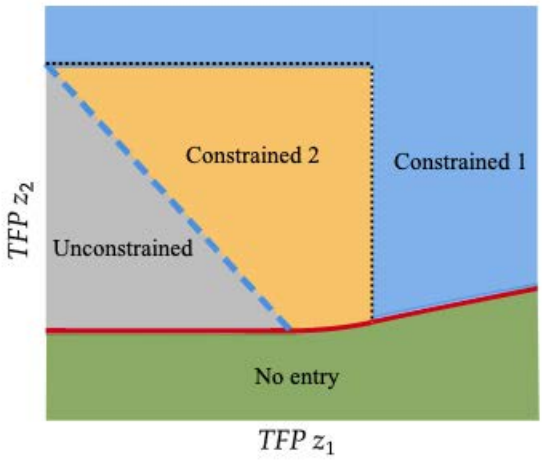
QED

Figure B-1 illustrates the decision to enter for a second firm. The graph shows the threshold $Z(z_1, e)$ for the TFP of the 2nd firm as a function of the TFP of the first firm z_1. If the TFP draws z_1 and z_2 are low relative to the entrepreneur's equity, she will be unconstrained in the sense that she has sufficient equity to fund both firms at the optimal size (area labeled "Unconstrained").[38] The threshold is therefore constant at $Z = z^*$.

For intermediate levels of z_1 and z_2 the entrepreneur will be constrained when operating two firms but unconstrained when operating one firm (area labeled "Constrained 2"). In this case the opportunity cost of equity is larger and this cost is increasing in z_1. Therefore, the threshold $Z(z_1, e)$ is monotone increasing in z_1. The dotted black line marks the combinations of z_1 and z_2 for which equity would be exactly sufficient to fund the most productive firm at the optimal size, $\lambda e = z_j k^*$.

For higher levels of z_1 and z_2 the entrepreneur will be constrained even when operating just one firm (area labeled "Constrained 1"). This further increases the opportunity cost of equity and the threshold keeps growing in z_1.

Figure B-1: Entry Decision for 2nd Firm.



Notes: The figure shows the entry threshold for the 2nd firm of entrepreneurs as a function of the TFP of the entrepreneur's first firm, z_1.

The opportunity cost of equity is lower when equity is more abundant. Entrepreneurs with more equity are therefore more likely to start the second firm. It follows that $Z(z_1, e)$ is monotone decreasing in e.

Note that $Z(z, e)$ is always below the 45-degree line in z. Since it was optimal to operate the first firm (with TFP z_1) in the first period, it must also be better to operate this firm in the second period than not operating any firms.[39] The fact that the birth date of each firm is irrelevant implies that the entrepreneur will always choose to operate the most productive firm in the second period. It follows that $z_2 > z_1$ is a sufficient condition for the second firms to be operated and for the entrepreneur to become a SE. This is why the threshold function satisfies $Z(z_1, e) \leq z_1$.

[38] The blue dashed line marks the combinations of z_1 and z_2 for which equity is exactly sufficient to fund both firms at the optimal size, $\lambda e = (z_1 + z_2)k^*$.

[39] The reason is that the wages and interest rates are assumed to be constant over time. Moreover, the entrepreneur's equity e must be at least as large as what the entrepreneur had available in the beginning of the first period – otherwise it would not have been optimal to operate the firm in the first period.

B.4 Proof of Proposition 4

From Proposition 3 the condition for choosing to operate firm 2 in period 2 is

$$\rho z_1 + \varepsilon - Z\left(z_1, e_2\right) \geq 0, \tag{B-9}$$

where Z is monotone increasing in z_1. By taking the partial differential of the functions $\underline{Z}$ and $\bar{Z}$ with respect to z_1 it is straightforward to show that $\underline{Z}$ is convex in z_1 while $\bar{Z}$ is concave in z_1,

$$\begin{aligned}
\frac{\partial^2 \bar{Z}}{\partial z^2} &= -X\left(1+\left(\frac{w}{\alpha\eta}\right)^{\frac{\alpha\eta}{1-\alpha\eta}}\frac{\nu}{1-\alpha\eta}\left(\lambda e\right)^{-\frac{(1-\alpha)\eta}{1-\alpha\eta}}\left(z\right)^{-\frac{1-\eta}{1-\alpha\eta}}\right)^{\frac{(1-\alpha)\eta}{1-\eta}-1}\left(\lambda e\right)^{-\frac{(1-\alpha)\eta}{1-\alpha\eta}}\left(z\right)^{-\frac{1-\eta}{1-\alpha\eta}-1} < 0 \\
\frac{\partial^2 \underline{Z}}{\partial z^2} &= \frac{\left(1-\alpha\right)\eta}{1-\alpha\eta}\frac{k^*}{\lambda e}\left(\frac{1-\eta}{1-\alpha\eta}\frac{zk^*}{\lambda e}+\frac{R\lambda e+\nu}{1-\alpha\eta}\frac{1-\alpha}{R}\frac{\eta}{\lambda e}\right)^{\frac{(1-\alpha)\eta}{1-\eta}-1} > 0.
\end{aligned}$$

where $X = \nu\left(1-\alpha\right)\eta/\left(1-\alpha\eta\right)^2\left[w/(\alpha\eta)\right]^{\alpha\eta/(1-\alpha\eta)} > 0$ is a constant. Since $Z\left(z,e\right)$ is convex in z for $z < \lambda e/k^*$ and concave in z for $z \geq \lambda e/k^*$, the largest value of $\partial Z\left(z,e\right)/\partial z$ occurs for $z = \lambda e/k^*$. It follows that $\partial Z\left(z,e\right)/\partial z$ is bounded from above by the following expression,

$$\frac{\partial Z\left(z,e\right)}{\partial z}\,|_{z=\lambda e/k^*} = \left(1+\frac{1-\eta}{1-\alpha\eta}\frac{z^*}{z}\right)^{\frac{(1-\alpha)\eta}{1-\eta}} - 1 < \left(1+\frac{1-\eta}{1-\alpha\eta}\right)^{\frac{(1-\alpha)\eta}{1-\eta}} - 1,$$

where the last inequality follows from the maintained assumption that $z_1 = \lambda e/k^* \geq z^*$. Recall now that equity $e_2 = \Pi(z_1, e_1)$ is given by the accumulated equity after operating the 1st-SE firm for one period. Since profits are monotone increasing in TFP, Assumption 1 guarantees that e_2 is monotone increasing in z_1. Equity e_2 therefore mitigates the degree to which Z is increasing in z_1. Therefore, the inequality (B-10) provides an upper bound for the derivative $dZ\left(z_1, \Pi(z_1, e_1)\right)/dz_1$. It follows that if ρ is sufficiently large, the expression $\rho z_1 + \varepsilon - Z(z_1, e_2)$ is monotone increasing in z_1. Lemma 1 therefore applies and implies that

$$E\{z_1 | z_1 \geq z^{**} \text{ and } \rho * z_1 + \varepsilon - Z\left(z_1, \Pi(z_1, e_1)\right) \geq 0\} \geq E\{z_1 | z_1 \geq z^{**}\},$$

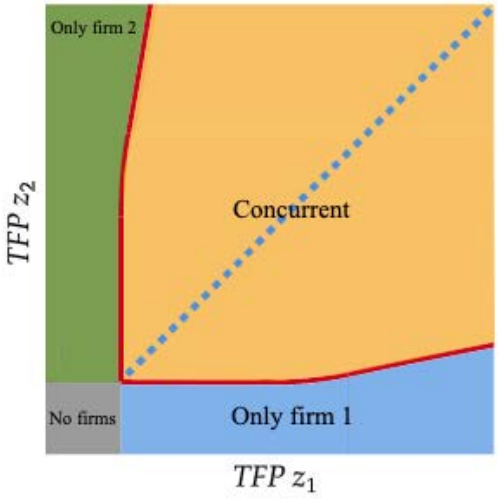
which establishes that the expected TFP of 1st-SE firms will exceed the expected TFP of non-serial entrepreneurs. Finally, following the proof of Proposition 2, a sufficiently large ρ guarantees that the expected TFP of the 2nd-SE firm will exceed the expected TFP of the 1st-SE firm.

B.5 Concurrent versus Non-concurrent SE Firms

The TFP difference $z_1 - z_2$ matters because the opportunity cost of operating the least productive firm is increasing in TFP of the most productive firm. Intuitively, if the TFP difference $|z_2 - z_1|$ is sufficiently large, it is optimal to allocate the entire endowment of the scarce factor to the most productive firm. Figure B-2 illustrates this aspect of Proposition 6. In a range close to the 45-degree line – when z_1 is close to z_2 – it is optimal to operate the firms concurrently. However, when the difference $|z_2 - z_1|$ is large (one firm being much more productive), the opportunity cost of equity becomes so large that it is optimal to allocate all funds to one firm and not operate the least productive one. In other words, the larger the difference in TFP, the lower the chance that the entrepreneur will operate both firms concurrently. In the same vein, when equity is more abundant, the opportunity cost of equity is lower. This explains why more equity increases

the chance that the entrepreneur will operate both firms concurrently.

Figure B-2: Entry Decision for 2nd Firm.



Notes: The figure illustrates the choice of whether to operate two firms concurrently or to operate just the most productive firm in the second period. These choices are determined by the combinations of z_1 and z_2 in the regions marked as "Concurrent", "1st firm only", or versus "2nd firm only".

B.6 Proof of Proposition 8

Consider two sectors with identical realizations of the idiosyncratic draws $z_{s'} = z_{\tilde{s}}$. Since the sector-specific return to capital has the same mean and variance in all sectors and $E\{u(W)\}$ is strictly falling in the covariance term, sector s' will be strictly preferred to sector $\tilde{s}$ if and only if $Cov(\delta_{s'}, \delta_s) < Cov(\delta_{\tilde{s}}, \delta_s)$. Since the distribution of δ is the same for all sectors, it follows immediately that when $Cov(\delta_{s'}, \delta_s) < Cov(\delta_{\tilde{s}}, \delta_s)$ then sector $\tilde{s}$ will be chosen only if it has the largest TFP, $z_{\tilde{s}} > z_{s'}$. This implies that 2nd-SE firms in sectors with a larger covariance with the sector of the 1st-SE firm will on average have a larger TFP. It follows that sector s' will be chosen more often than sector $\tilde{s}$.

C Favored and Non-Favored Entrepreneurs with Sector Learning

Consider an extension of our learning model where favored entrepreneurs (i.e., individuals who possess non-skill advantages) coexist with non-favored entrepreneurs. For ease of exposition, assume that the advantage of the favored entrepreneurs takes the form that they do not face strict collateral constraints (i.e., very large λ). Since favored entrepreneurs can borrow more, they have a lower TFP threshold z^* than the non-favored entrepreneurs: $z^*_{nf}(e) \geq z^*_f(e)$.

A key implication of this setting is that favored individuals will be more preponderant among the sector switchers than among the stayers. To understand this result, note that there are no differences between the favored and the non-favored in their second-period decision rule on switching sectors because all entrepreneurs have the same switching threshold $\bar{z}$. Consider the case when $\bar{z} > z^*_f(e)$, which is necessary to observe some entrepreneurs switch sector in equilibrium. Otherwise, all entrants would have TFP above the switching threshold and would choose to remain in their initial sector. In this case, favored entrepreneurs would be more over-represented among the switchers than among the stayers. Since favored entrepreneurs have a lower TFP threshold for entry than the non-favored ones (i.e., $z^*_f(e) \leq z^*_{nf}(e)$), the favored should be proportionally represented among entrepreneurs who choose to enter for sufficiently high realizations of z and over-represented for low realizations of z. This holds true for any level of equity e. It follows that the preponderance of favored entrepreneurs should be larger for TFP levels below the switching threshold $\bar{z}$ than for TFP levels above this threshold.

This result allows us to use sector switching as a proxy for favored individuals.

An implication of this result is that the presence of favored individuals will be a force for lower TFP and larger capital stock among switchers. In particular, switchers will have more capital than stayers, relative to their TFP. If the advantage is sufficiently large, the switchers could end up having larger average capital than the stayers, even if their average TFP is lower.

We summarize these insights in the following corollary.

Corollary 2. *Favored entrepreneurs are over-represented among entrepreneurs who switch and locate the 1st-SE and 2nd-SE in different sectors. Switchers therefore have more capital relative to their TFP than stayers. If the favored entrepreneurs enjoy sufficiently large advantages in terms of borrowing, switchers could have larger average capital stocks than stayers.*

In this section we have focused on easier access to borrowing (i.e., a larger λ) as the source of favoritism. The effects of favoritism would be quantitatively stronger a lower cost of borrowing (i.e., a lower interest rate).

D Industry-Specific Labor Income Shares

Following much of the literature on firm-level productivity measurements in China (including for example Brandt et al. (2012), Hsieh and Klenow (2009), and Bai, Hsieh and Qian (2006)), our preferred approach when estimating firm-level TFP is to use sector-specific labor-income shares from the U.S. The values for sector-specific labor-income shares for the U.S. are reported in column 1 of Table D-1 (source: BEA).

Unfortunately, it is not possible to estimate sector-specific labor-income shares using the Inspection Data, which our analysis is based on. The reason is that these data do not include information on labor and wages. We estimate the labor-income share from Chinese data in two ways for each sector j – as a pure labor-income share $\phi_{1,j}$ and as an elasticity $\phi_{2,j}$.

We measure the share parameter $\phi_{1,j}$ as the ratio of aggregate labor income to aggregate value added in industry j. We rely on data from the Chinese 2007 Input-Output table to measure the labor compensation and value added for each sector. The Chinese 2007 Input-Output table also includes each industries' use of labor, capital depreciation, surplus, and paid-in tax. We aggregate data at the 3-digit industry level up to a 1-digit industry level, and use this to calculate the labor share ratio of 1-digit industries. When aggregating up to the 1-digit level, the weight for each 3-digit industry is the number of firms from the 2008 census data.

We measure the parameters $\phi_{2,j}$ as the elasticity of value added to a change in labor input in industry j. Formally, we estimate $\phi_{2,j}$ running the following OLS regression using data from the 2008 Chinese Enterprise Census Data.

$$\ln y_{i,j} = (0.85 - \alpha_j) \ln(k_{i,j}) + \alpha_j \ln(n_{i,j}) + \varepsilon_{i,j} \tag{D-1}$$

where $y_{i,j}$ is value added for firm i in industry j, $k_{i,j}$ is capital, and $n_{i,j}$ is the labor input. The decreasing returns to scale parameter of 0.85 is taken from Restuccia and Rogerson (2008).

Table D-1 reports the results. We have two main findings. First, the measured labor share estimates are positively correlated with the corresponding shares for U.S. industries, with a correlation of about 0.3. We conclude that the U.S. labor shares are informative about labor elasticities in Chinese data.

Second, we observe that for the largest industries (i.e., Wholesale and Retail, Manufacturing, and Leasing and Business Services), the estimated labor shares using Chinese data are low compared to their U.S. counterparts (see columns 2 and 3 of Table D-1). Manufacturing is a point in case, where the estimates for China are around 0.33. This compares to 0.514 for the U.S. It seems unreasonable to us that China should have a substantially lower aggregate labor share in manufacturing than the U.S. This could be due to the fact that firm-level data probably do not include all labor compensation because contributions through pension plans, insurance, etc. might be ignored (this issue is relevant also when estimating labor shares in U.S. data). Furthermore, Chinese data are likely to mis-measure labor compensation because some firms report costs of hired temporary workers as "intermediary inputs" rather than as part of the labor cost. Finally, the measured production parameters might be biased because of sector-specific distortions, including subsidies or taxes (see for example Hsieh and Klenow (2009) and Brandt et al. (forthcoming)).

Therefore, we find it reasonable to rely on U.S. labor-income shares when estimating firm-level TFP. The reason is that these measurement issues are arguably much smaller in the U.S.

Table D-1: Labor Share Estimation

1-digit industry	ϕ_j^{US}	$\phi_{1,j}$	$\phi_{2,j}$
	(1)	(2)	(3)
Mining	0.264	0.352	0.254
Manufacturing	0.514	0.324	0.338
Utilities	0.269	0.254	0.377
Construction	0.641	0.510	0.454
Wholesale and Retail	0.533	0.242	0.233
Transportation	0.623	0.271	0.420
Catering	0.580	0.276	0.613
IT	0.399	0.230	0.328
Finance	0.238	0.260	0.551
Real Estate	0.238	0.109	0.316
Leasing and Business Service	0.714	0.552	0.488
R&D	0.387	0.537	0.387
Public Facilities	0.387	0.384	0.453
Resident Service	0.387	0.679	0.379
Education	0.387	0.663	0.380
Health	0.580	0.509	0.398
Culture	0.580	0.820	0.451
Social Organization	0.413	0.887	0.314
International Organization	0.413	–	0.451
$CORR(\phi_j, \phi_j^{US})$	–	0.266	0.306

Notes: ϕ_j^{US} is from the BEA; $\phi_{1,j}$ is from the Chinese Input-Output Table 2007; and $\phi_{2,j}$ is estimated based on the Chinese Enterprise Census Data 2008.